

Introduction to Earth System

Ocean

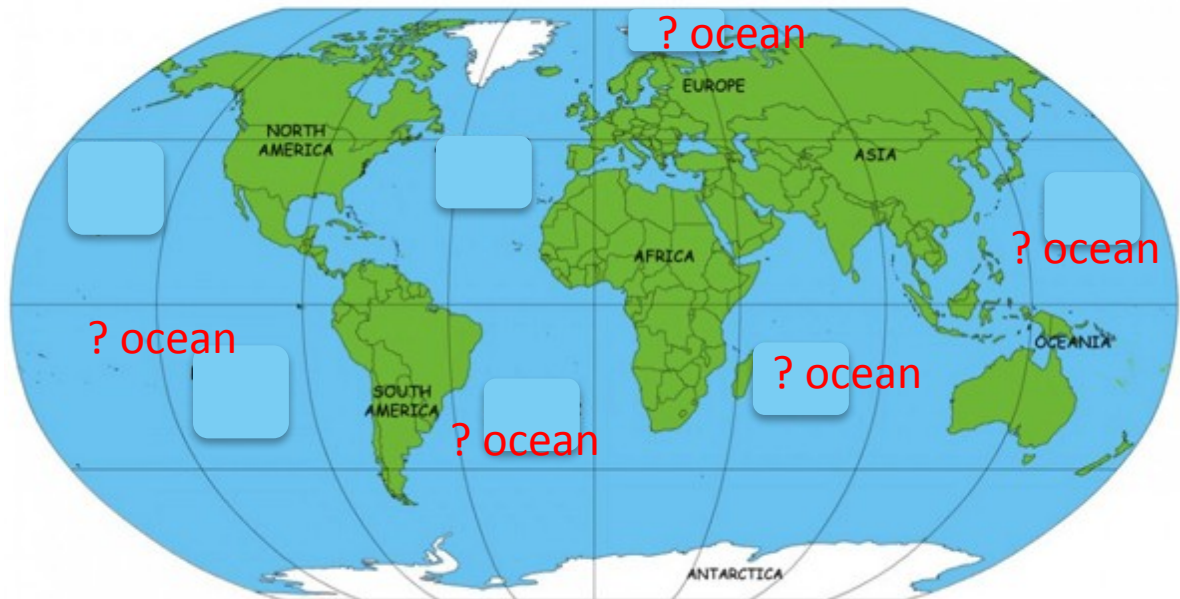
Ocean

1. Intro & Ocean structures
2. Ocean circulation
3. Ocean temperature
4. Sea level

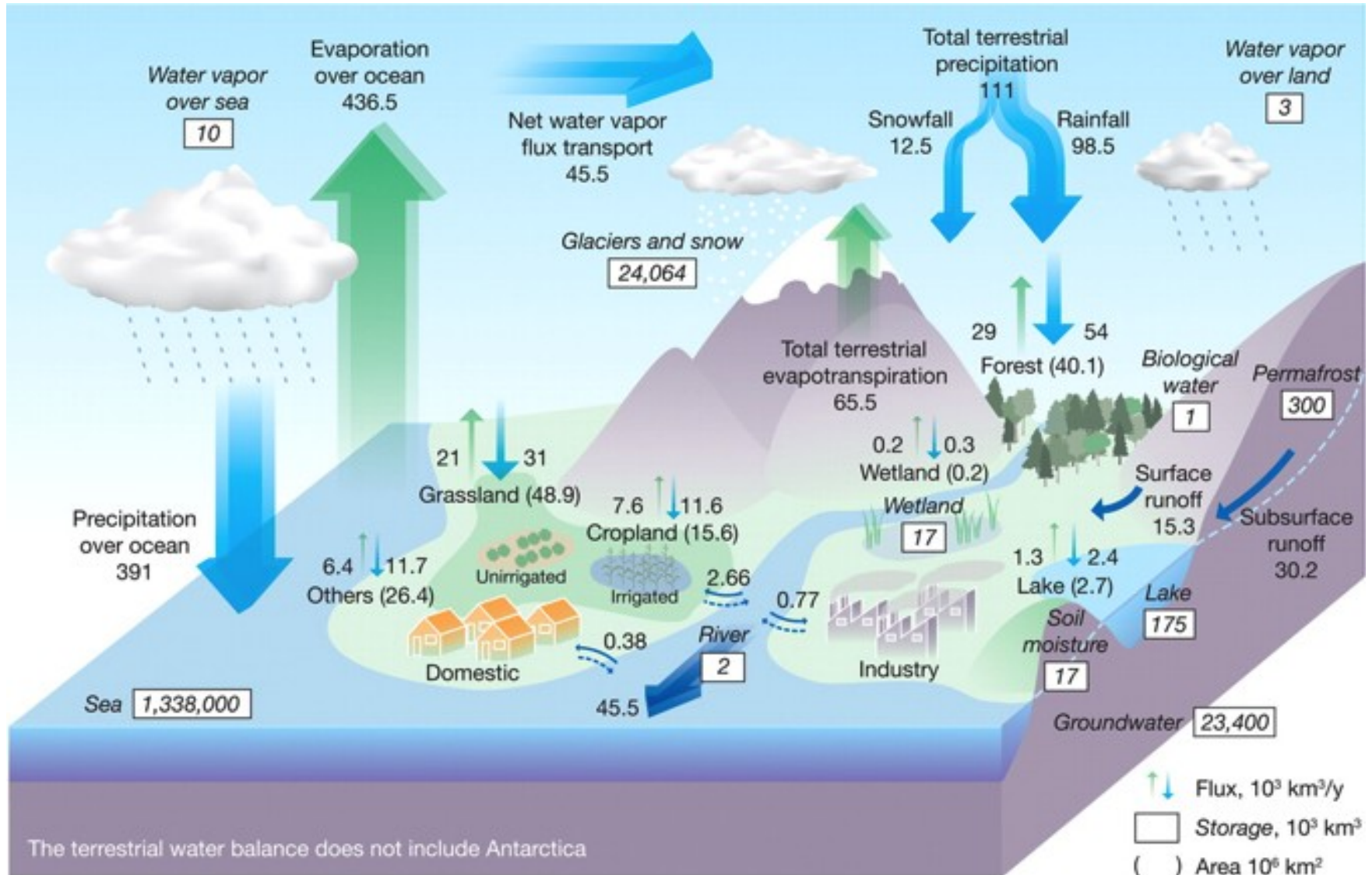
The Oceans

- Cover $Q\%$ of the area of the Earth's surface
- Reach an extreme **depth** of $\sim Q$ km
- **Total volume** is equivalent to that of a layer Q km deep, covering the entire surface of the Earth
- **Oceans' mass**: $\sim Q$ times compared to that of the atm.

OCEANS OF THE WORLD

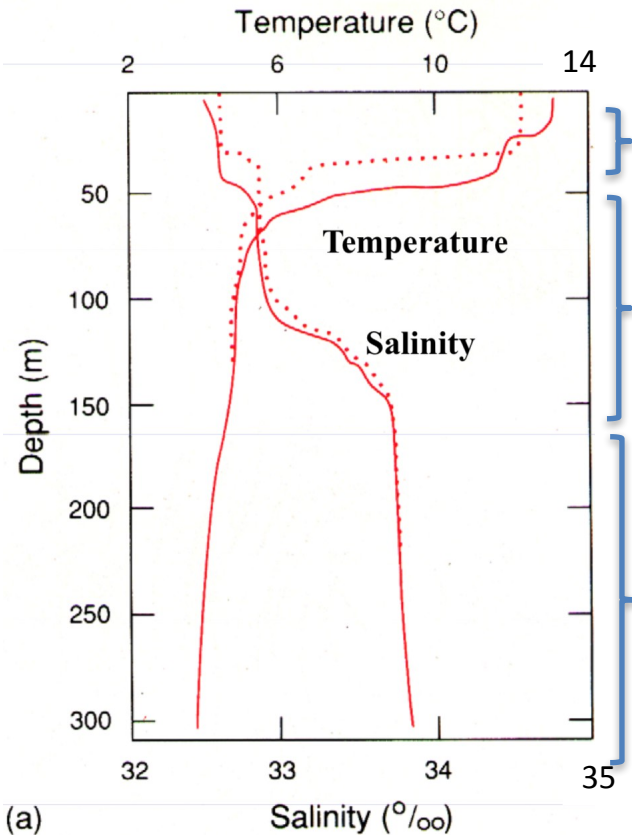


The water cycle



Ocean structures

- **The ocean is mostly stably stratified**
- **Upper ocean:** shallow (~100m), warm, abundant light, most marine life
- **Deep ocean:** cold, dark, much nutrients and carbon



(Climate system Modeling)

Mixed layer: T & S well mixed

- Atmosphere affect: heating, wind forcing & freshening (P-E)
- Global average depth ~ 70m
- Heat capacity of mixed layer → ~ 30 times heat capacity of the atmosphere

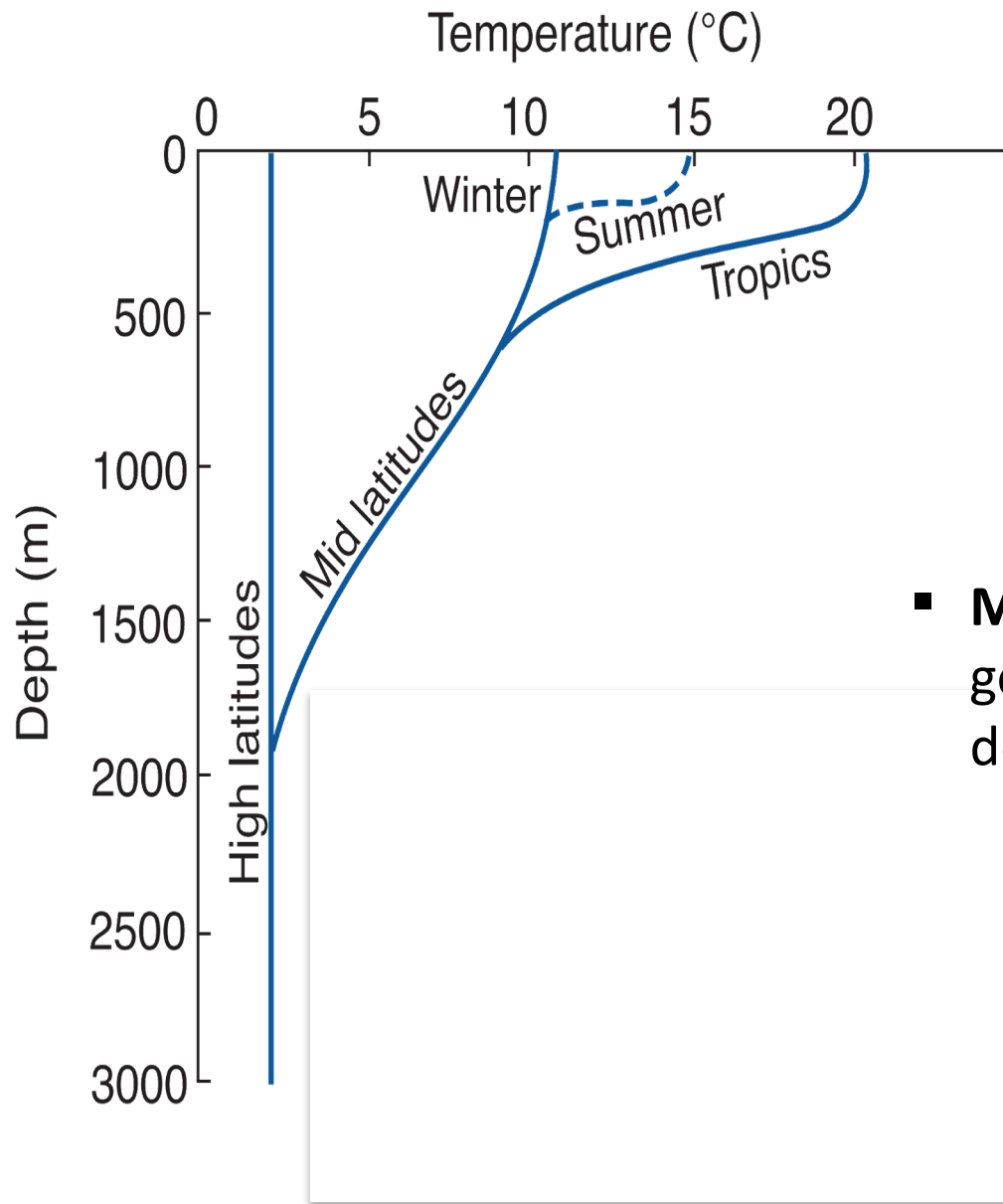
Thermocline: large gradient of T, S

Deep ocean: T, S ~independent of height

Q: Is mixed layer seasonally dependent?

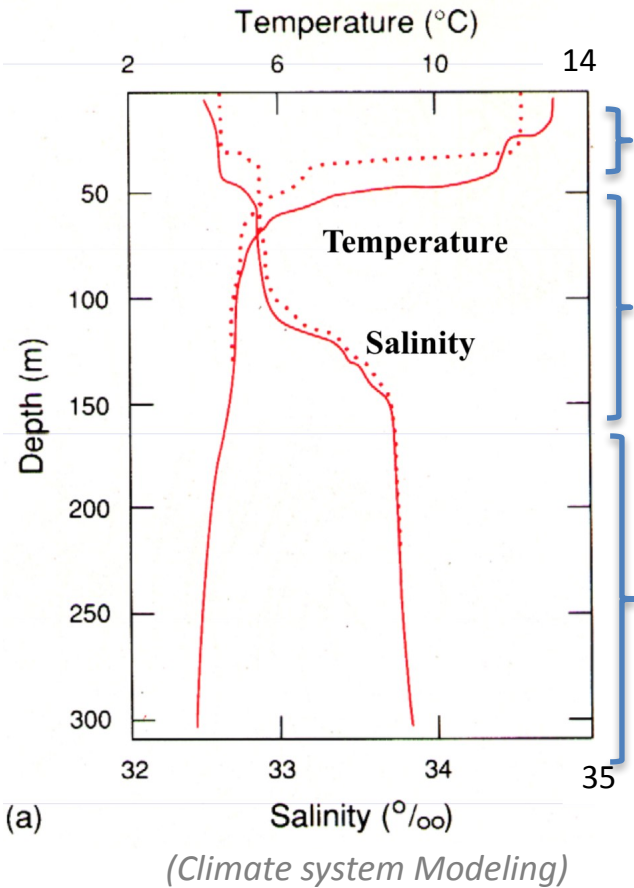
Q: Is mixed layer geographically dependent?

Ocean structures



- **Mixed layer:** seasonal & geographical location dependence

Ocean structures



Mixed layer

Thermocline: large gradient of T, S

*A **thermocline** is the **transition layer** between the warmer mixed water at the surface and the cooler deep water below, where the **temperature drops rapidly** with depth.*

Deep ocean

Exercise #1

An Automated Underwater Vehicle (AUV) dives vertically into the subtropical Atlantic Ocean and revealed the following temperature profile:

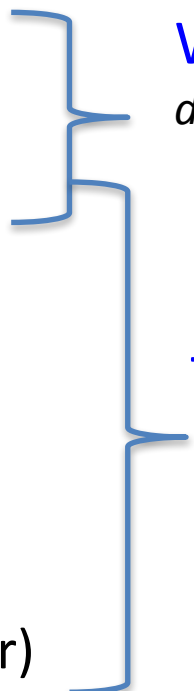
$$T(z) = 4 + \frac{22}{1 + e^{0.02(z-150)}}$$

- Estimate Temperature at the surface and in the deep ocean ($z \rightarrow \infty$)
- Locate the Thermocline depth
- Estimate the thermal gradient intensity at the thermocline location

Ocean circulation

Drivers:

- Wind stress on the surface
- Earth's rotation (Coriolis effect)
- Pressure gradients from density differences
- Buoyancy fluxes (heat, freshwater)



Wind-driven circulation
dominates the surface currents

The diagram uses blue curly braces to group the drivers. The first two drivers, 'Wind stress on the surface' and 'Earth's rotation (Coriolis effect)', are grouped by a brace pointing to the 'Wind-driven circulation' text. The next two drivers, 'Pressure gradients from density differences' and 'Buoyancy fluxes (heat, freshwater)', are grouped by a larger brace pointing to the 'Thermohaline circulation' text.

Thermohaline circulation
*density driven component;
circulation deeper in the oceans*

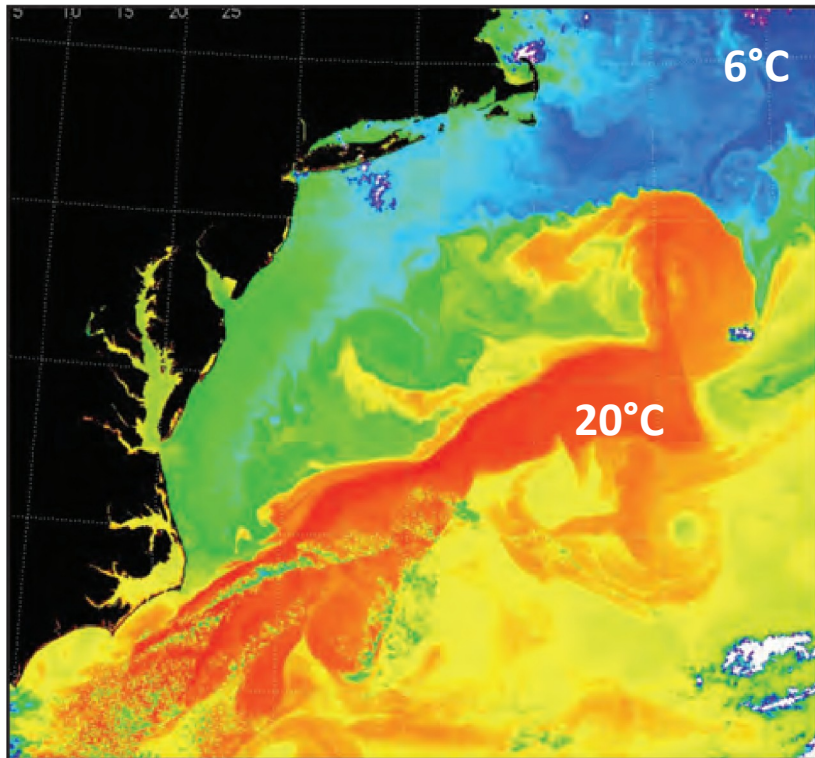
Fundamental Equations

1. Momentum (Navier-Stokes in rotating frame)
2. Continuity (mass conservation)
3. Thermodynamics (T, S conservation)
4. Equation of state ($\rho = \rho(T, S, p)$)

Ocean circulation

Wind-driven circulation:

- **Surface winds** → horizontal momentum from the atm. into the oceans → generating currents
- **Wind-driven currents** are typically on the order of 10 cm/s → a few % of surface wind speed

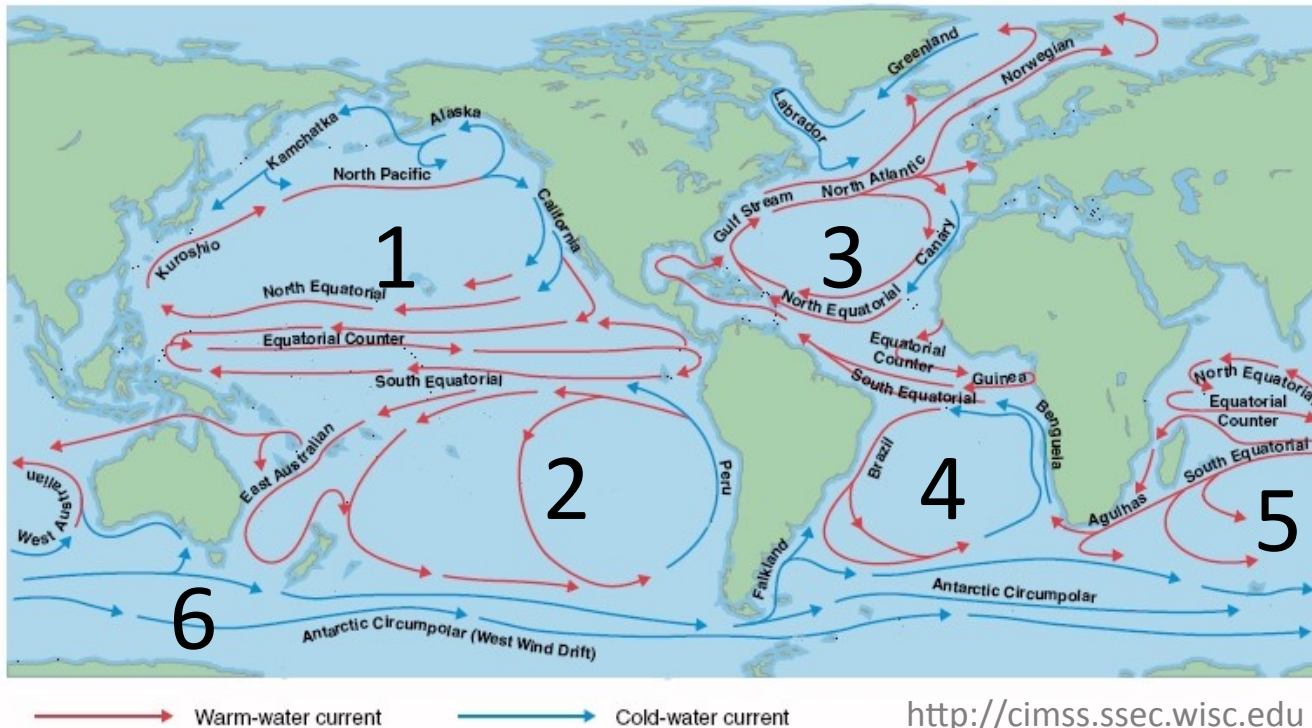


Gulf stream:

- Speed: 2m/s
- Depth: 450 m
- Width: 70 km

Eddies along the landward edge of the Gulf Stream

Six largest currents



5 geostrophic gyres:

1. North Pacific Gyre
2. South Pacific Gyre
3. North Atlantic Gyre
4. South Atlantic Gyre
5. Indian Ocean Gyre

&

6. The largest current:
Antarctic Circumpolar
Current (West wind
drift)

Gyres' characteristics: each gyre includes 4 current components

Two boundary currents:

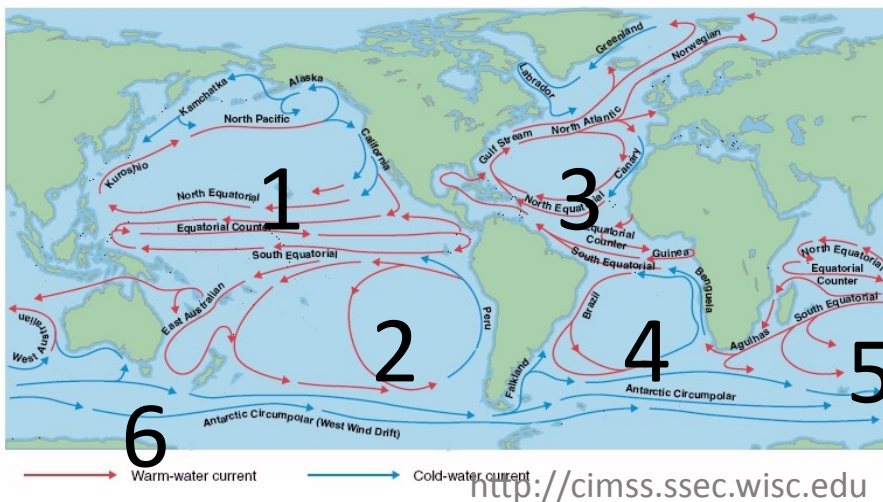
- western: fast, deep, narrow; moves warm water poleward
- eastern: slow, shallow, broad; moves cold water equatorward

Two transverse currents:

- Trade wind-driven: Moderately shallow & broad westward current
- Westerly-driven: wider & slower eastward current compared to the trade wind-driven current

Q: Why clockwise (anti-clockwise) gyres in the NH (SH)?

Exercise #2



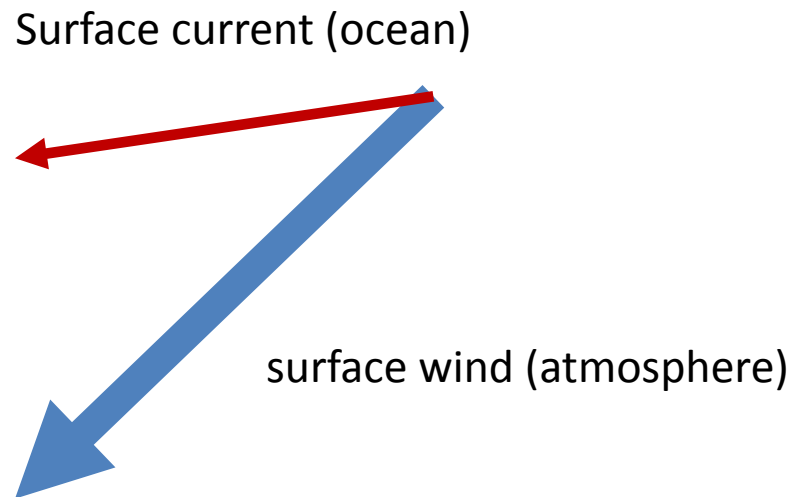
Name	Average width (W)	Average depth (D)	Average velocity	Average temperature
Gulf stream	80 km	800 m	1.2 m/s	22°C
Antarctic Circumpolar (ACC)	1000 km	2000 m	0.07 m/s	3°C

Unit of Ocean Volume Transport: 1 Sverdrup (Sv)= 10^6 m³/s

- Calculate the volume transport in Sv for the Gulf Stream and the ACC?
- Given density of seawater $\rho=1025$ kg/m³ and the specific heat capacity of seawater $C_p=4000$ J/(kg.°C). Determine the total heat energy flux the Gulf stream transports northward in Petawatts? (1 PW= 10^{15} W)

Surface winds & ocean currents

- Starting with a surface wind, e.g. the Northeast trade winds

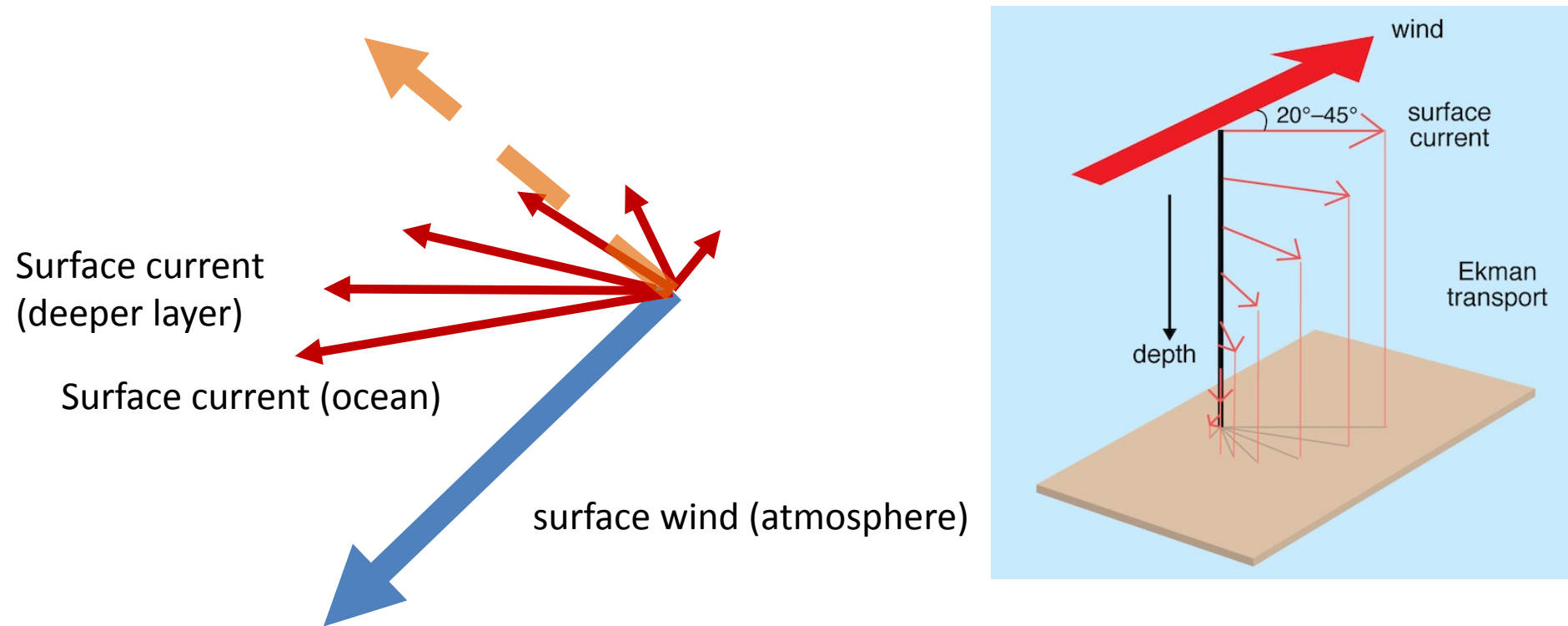


→ the ocean's surface layer current is:

- more slowly
- to the **right** of the wind

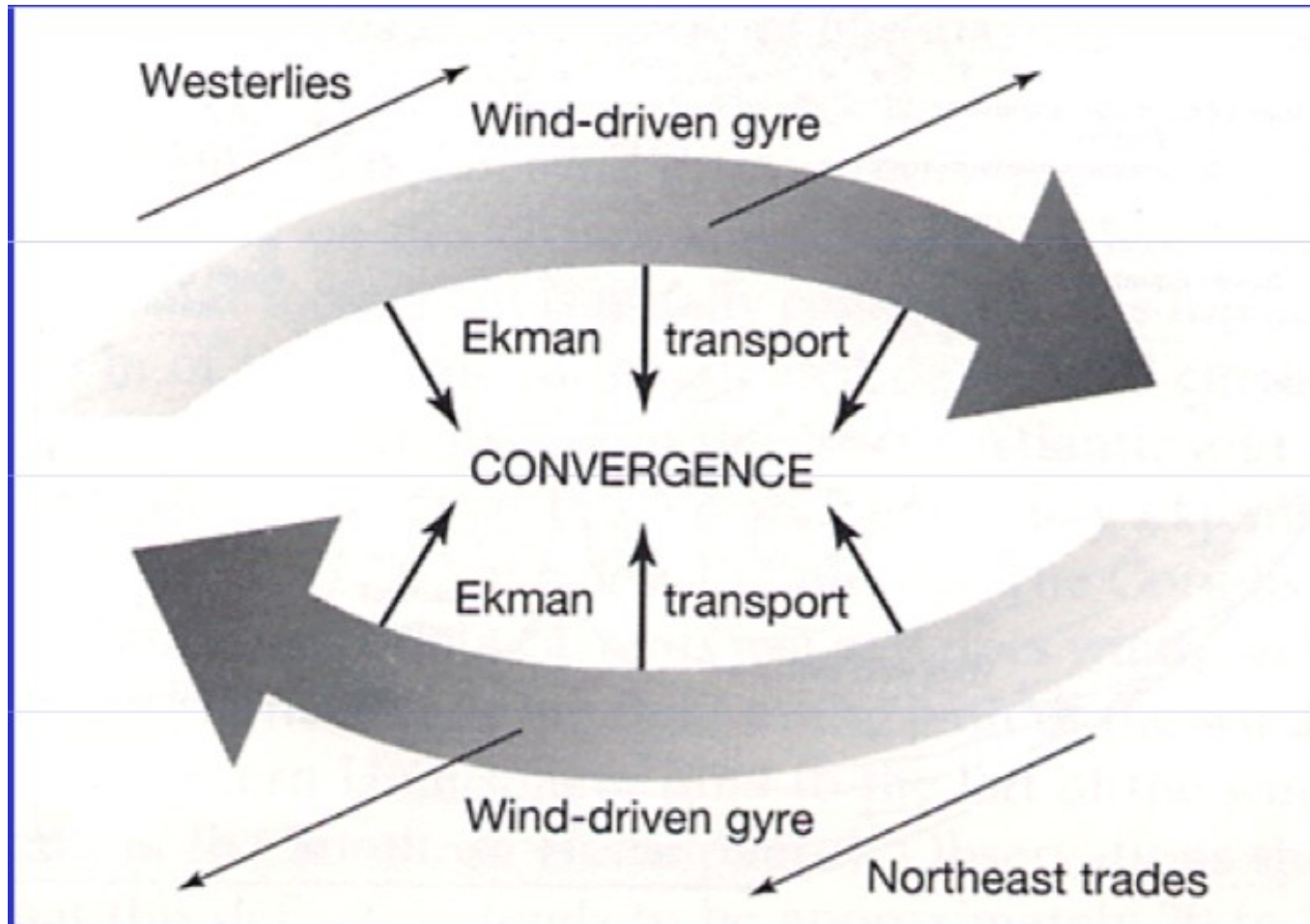
→ Q: Why?

Ekman transport



- Keep going down $\rightarrow$ the current both slowing and turning clockwise $\rightarrow$ an **EKMAN SPIRAL**
- Average the current and mass transport $\rightarrow$ the mean direction is to the right ($\sim 90^{\circ}$) of the surface direction $\rightarrow$ This mass motion is called the **EKMAN TRANSPORT**

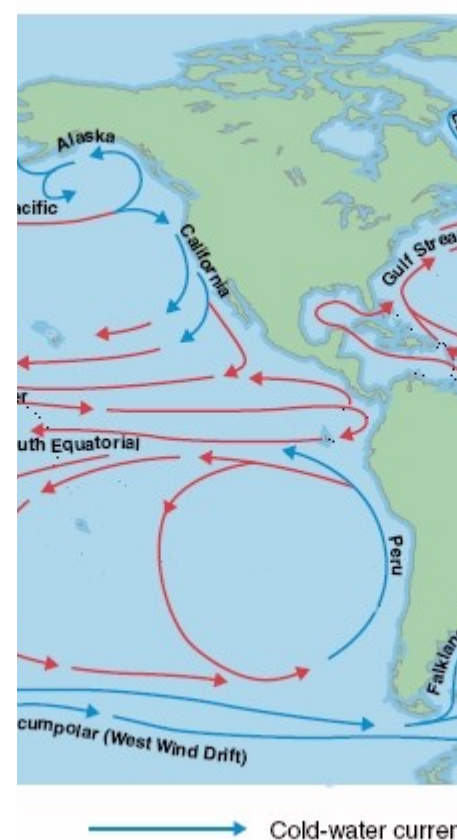
Ekman transport



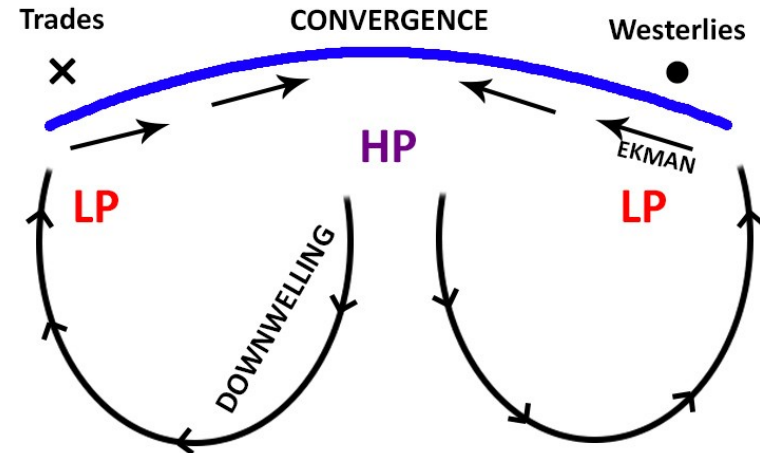
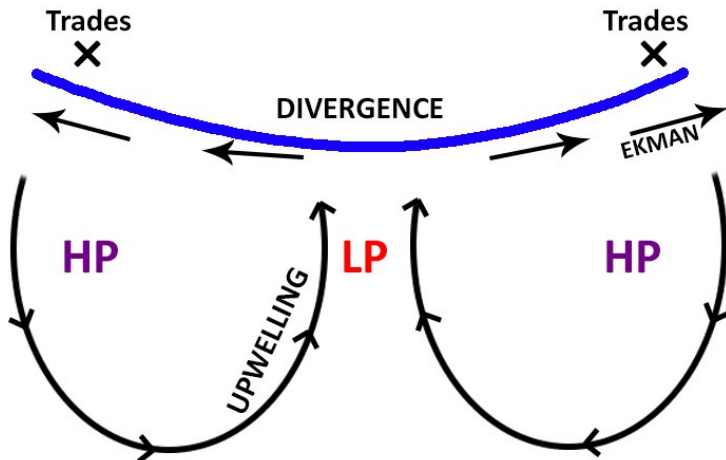
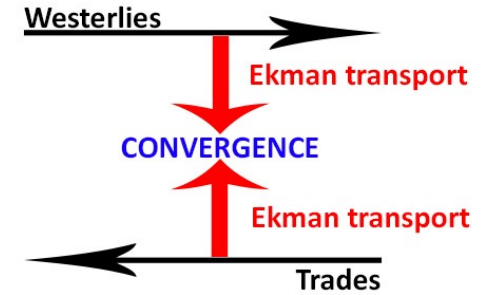
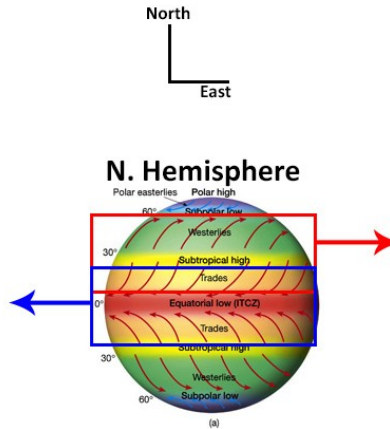
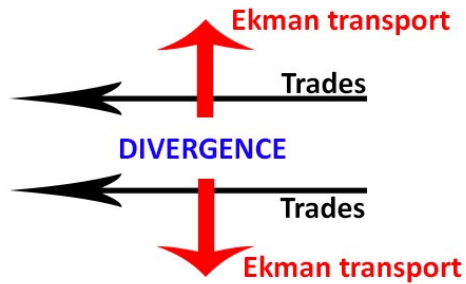
Ekman transport

E.g. west side of the American continent

- average movement of upper ocean water is westward, **AWAY** from shore.
- Push cool water away from the shoreline → colder water from the bottom is brought up → UPWELLING
- UPWELLING → transports nutrients to the surface (& cooler SST) → allowing phytoplankton to prosper

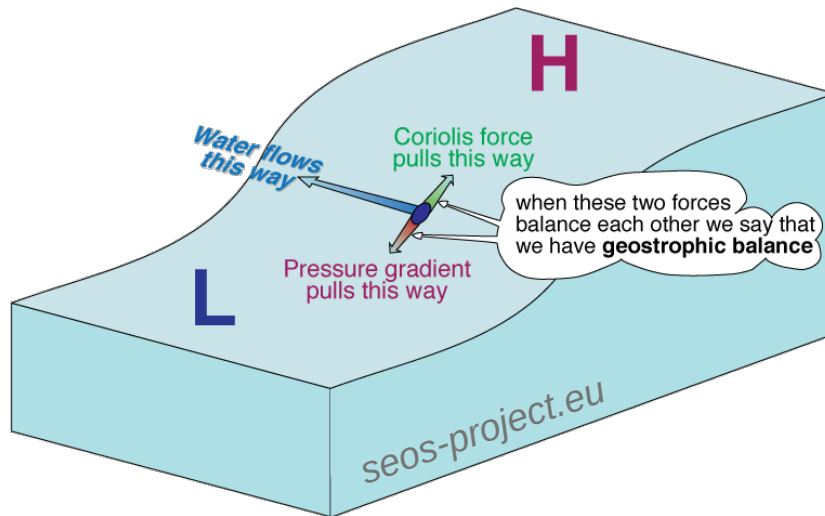


Upwelling & Downwelling



@C. Cameron

Ekman transport to geostrophic flow



$$-fv + \frac{1}{\rho} \frac{\partial p}{\partial x} = 0$$

$$fu + \frac{1}{\rho} \frac{\partial p}{\partial y} = 0$$

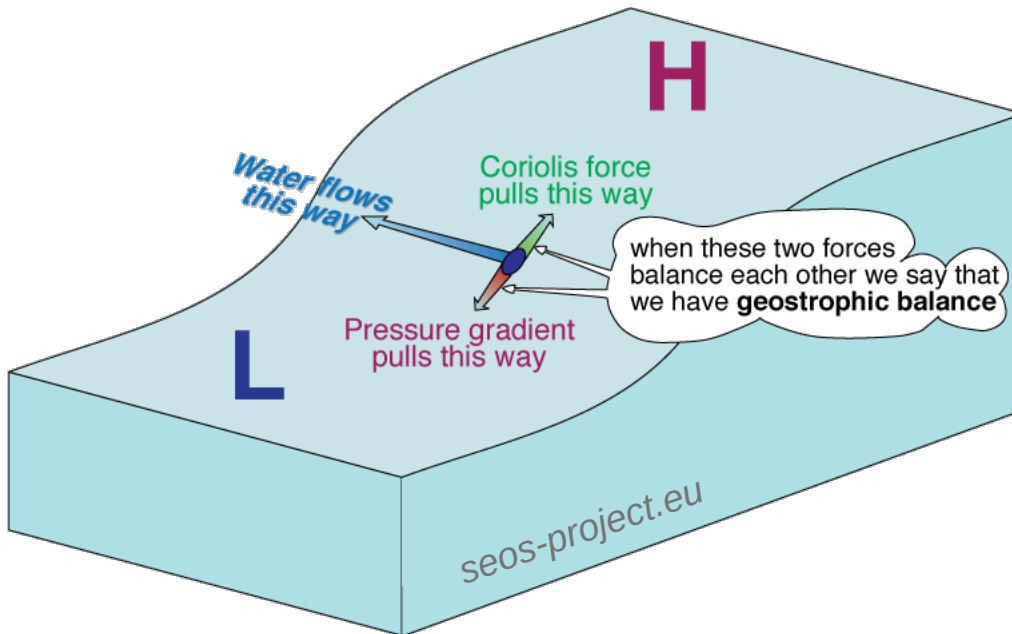
Mechanism

1. Winds → Ekman transport.
 - convergence (or divergence).
 - high (low) pressure zones in deep waters.
 - horizontal pressure gradient forces.
 - water to move horizontally from high to low pressure.
1. This horizontal water movement adjusts until it is 90° to the Coriolis effect and traveling at 90° to the pressure gradient force.

The Coriolis effect and pressure gradient force are equal but opposite → **geostrophic flow**

Geostrophic current

- pressure gradient is balanced by the Coriolis force

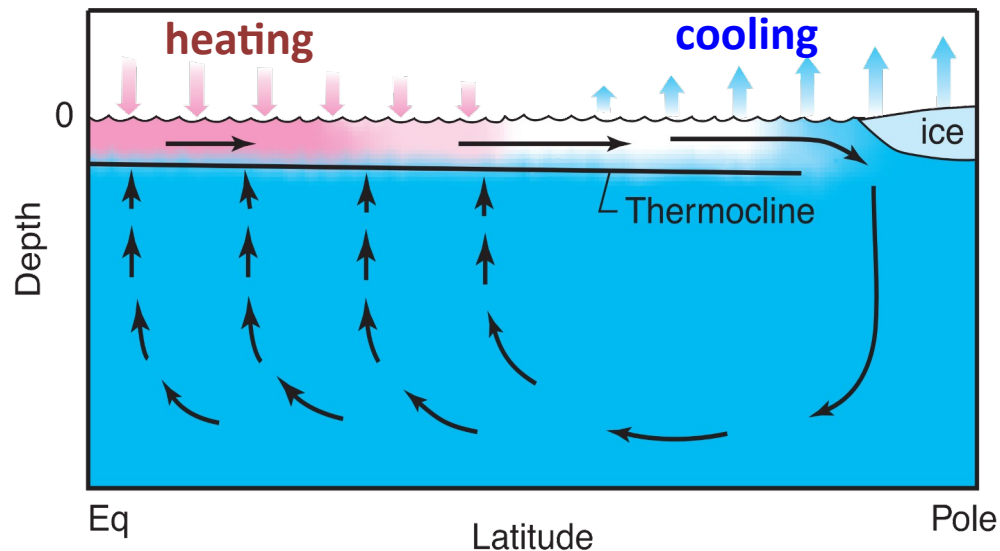


- The direction of geostrophic flow is parallel to the isobars
- In NH (SH), the high pressure is on the right (left) of the flow

- In the NH, the Coriolis force acts at right angles to the flow → clockwise gyre

Ocean circulation

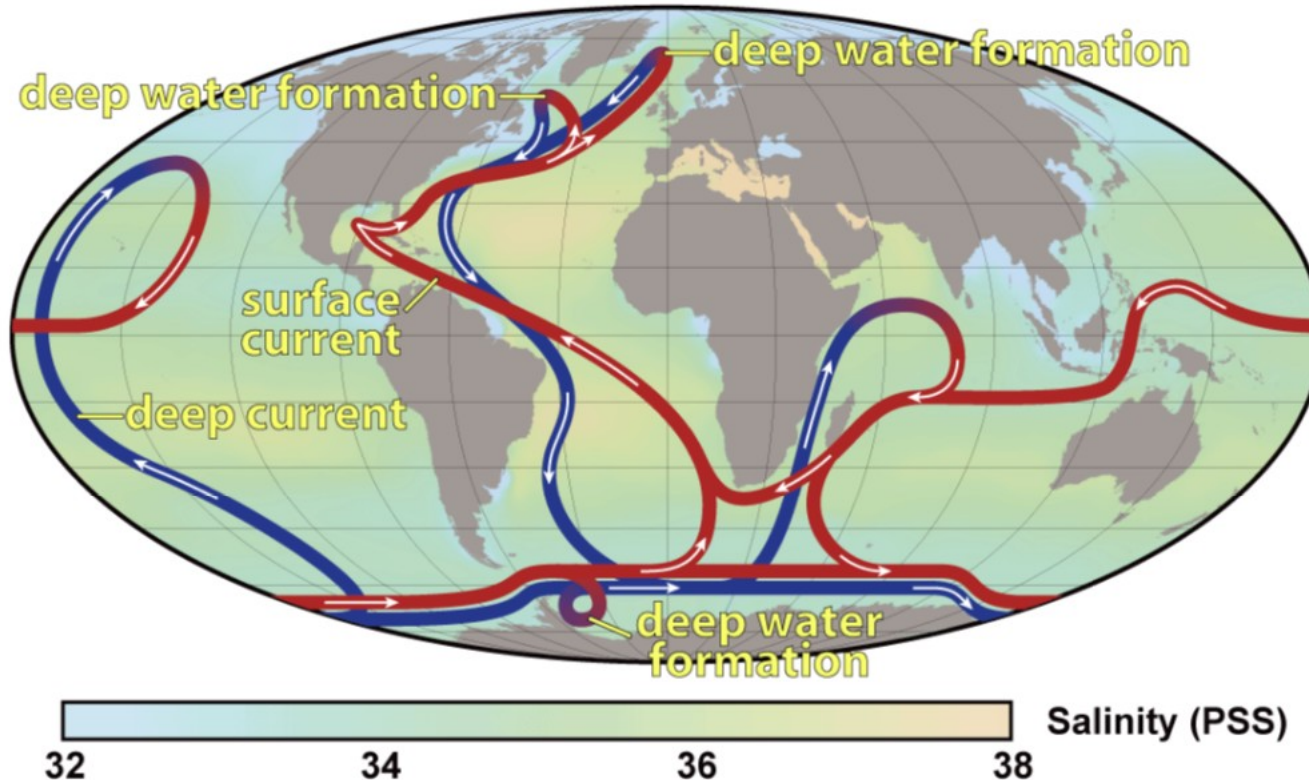
The idealized schematic of the thermohaline circulation



- Pink (blue) shading indicates warmer (colder) water
- Rejection of salt when water freezes along the ice edge that makes the **water dense** enough to enable it to sink to the bottom
- Timescale: relatively slow process, ~ hundred years

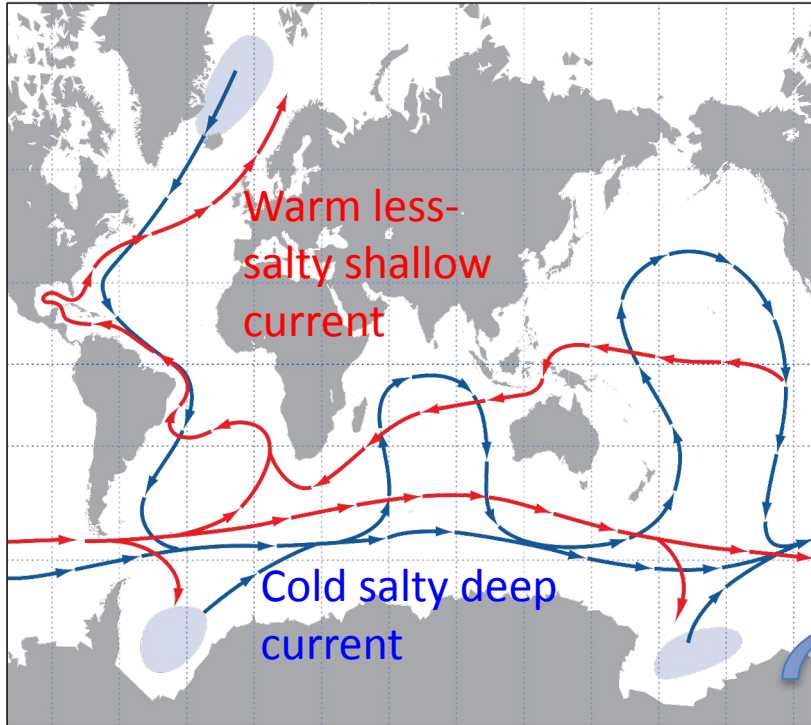
Ocean circulation

Thermohaline Circulation



- Although slow, a large amount of upper ocean water converted into deep bottom water, ~ **15,000,000 m³/s** in the Atlantic ocean.
- As a comparison, all the rivers in the world combined deliver ~ **1,000,000 m³/s** of fresh water into the ocean.

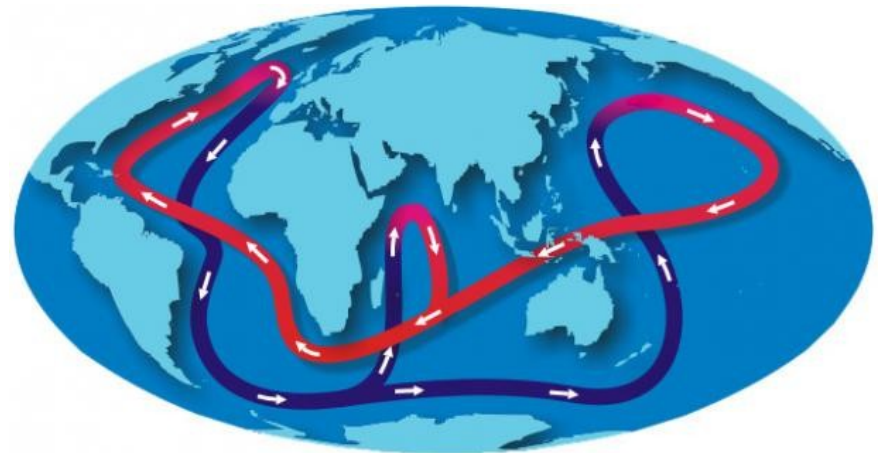
Ocean circulation



Simplified schematic of the thermohaline circulation.

- Shading → regions of downwelling
- Blue arrows: transport of bottom water
- Red arrows: return flow of surface water

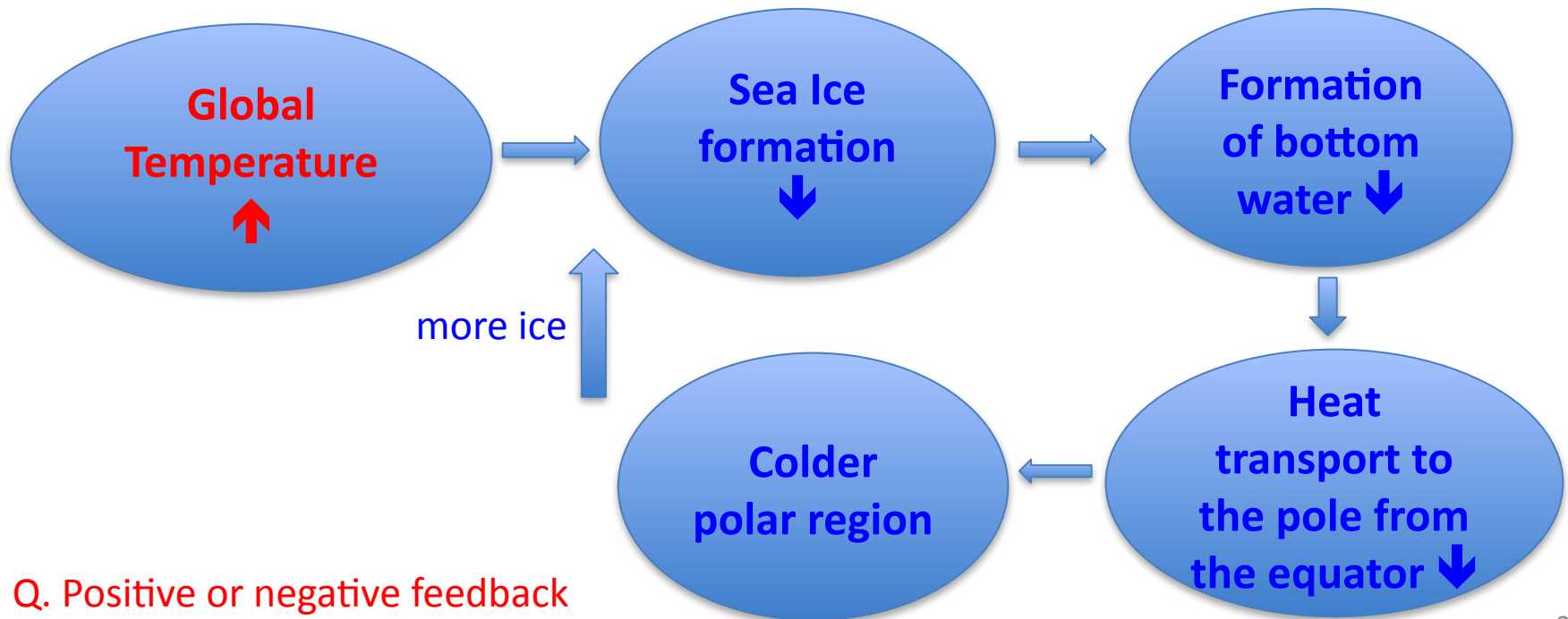
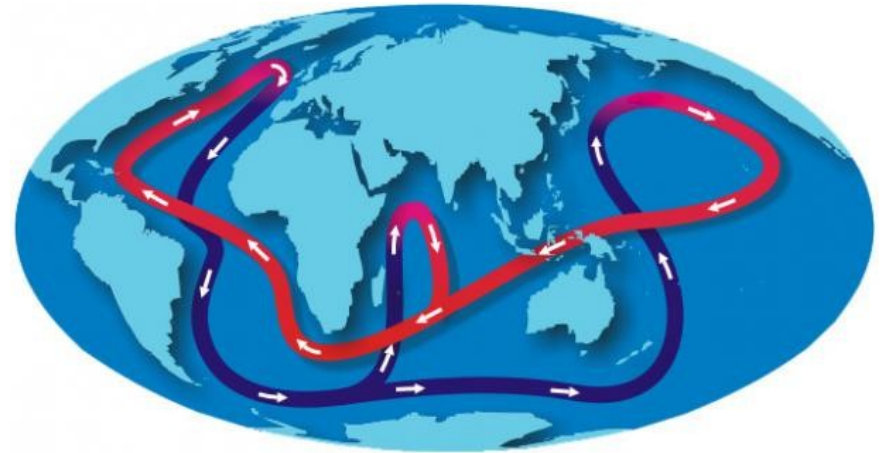
A very simplified model:
the Global Conveyor Belt



Ocean circulation

Climate Change & Global Thermohaline circulation

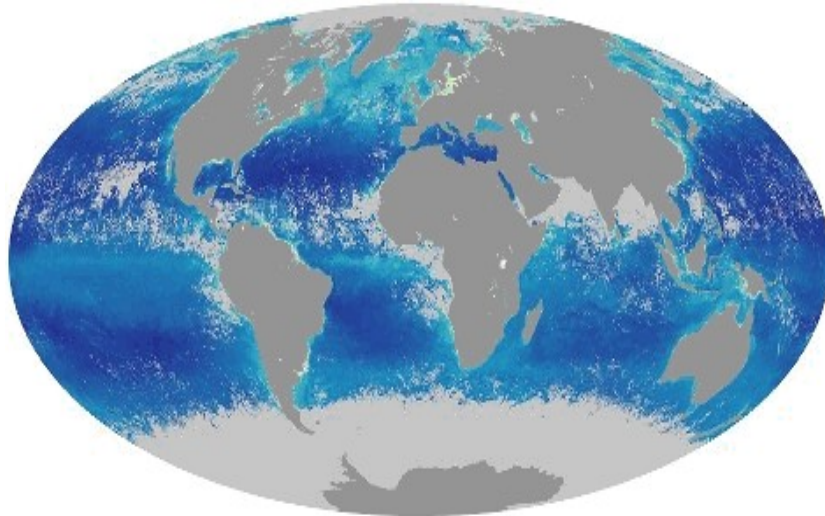
- Climate change studies suggest:



Q. Positive or negative feedback process?

Ocean circulation: Upwelling

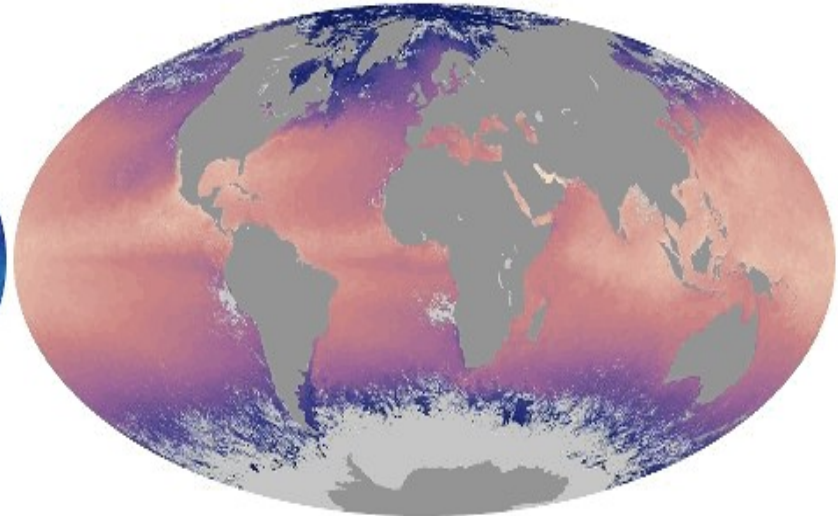
- **Upwelling:** the rise of **cold water** from deeper layers to replace warmer surface water
 - stimulate the growth & reproduction of phytoplankton
 - **~25% of the total global marine fish** catches come from five upwellings that occupy only **5% of the total ocean area**
- **Q: How to identify upwelling?**
 - Identified by **cool SST** & high concentration of **chlorophyll**



Chlorophyll Concentration
(mg/m³)

0.01 0.03 0.1 0.3 1 3 10 30 60

July 2002

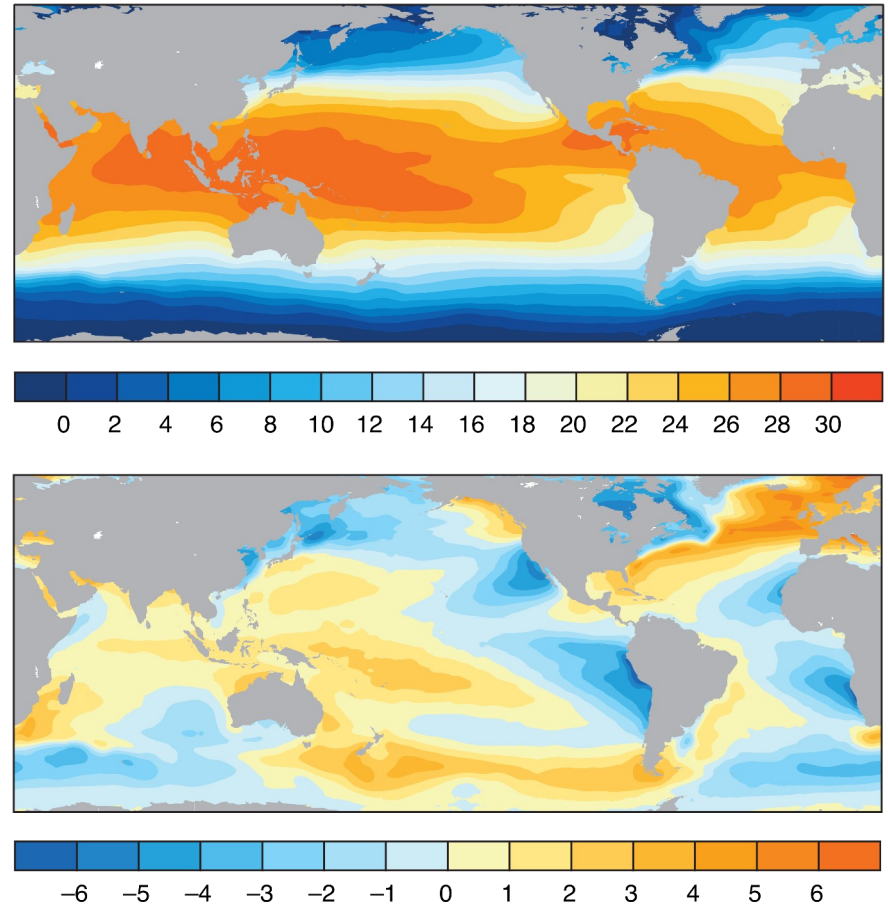


Sea Surface Temperature
°C

-2 35

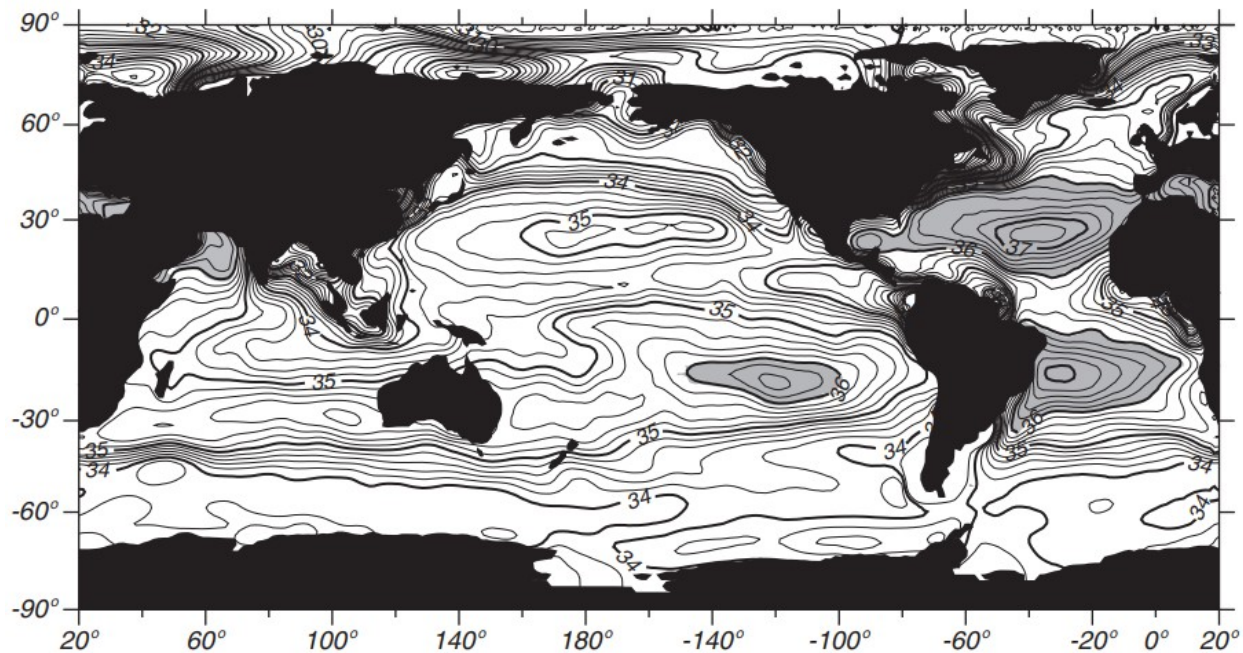
E.g. Sea Surface Temperature (SST)

- Radiative heating (e.g. north-south gradient, top fig.)
- Dynamical factors: (e.g. bottom fig.)
 - Subtropical latitudes
 - Higher latitudes
 - Equatorial regions

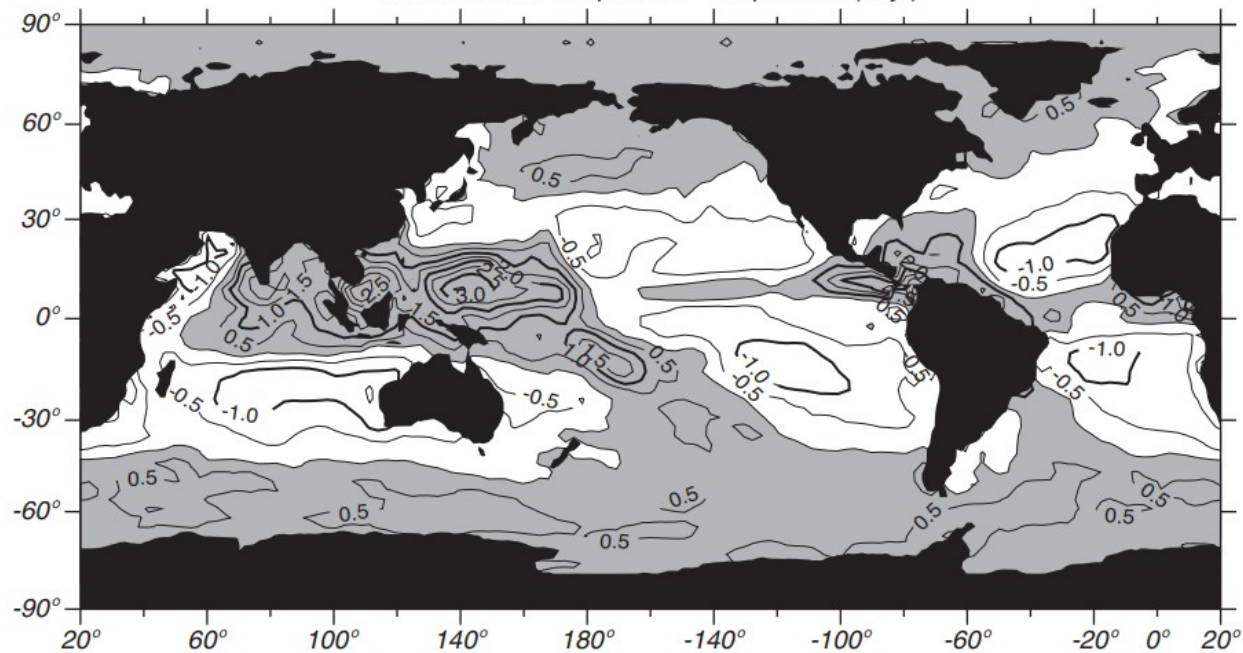


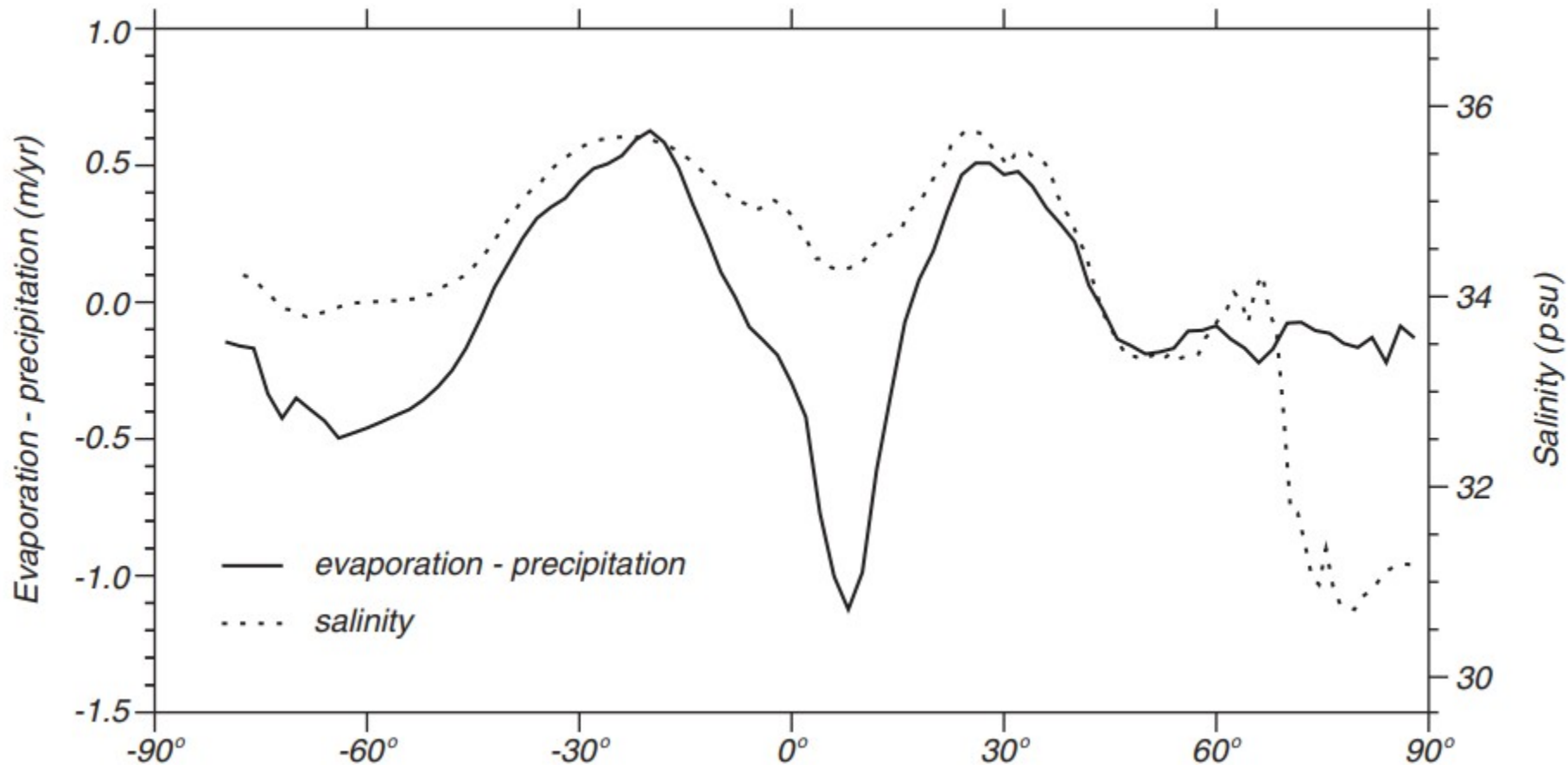
Annual mean SST. (Top) The total field
(Bottom) Departure of the local SST at each location from the zonal average field

Annual Mean Sea Surface Salinity



Annual Mean Precipitation – Evaporation (m/yr)





Zonal average of sea-surface salinity calculated for all the ocean from Levitus (1982) and the difference between evaporation and precipitation

Heat Capacity

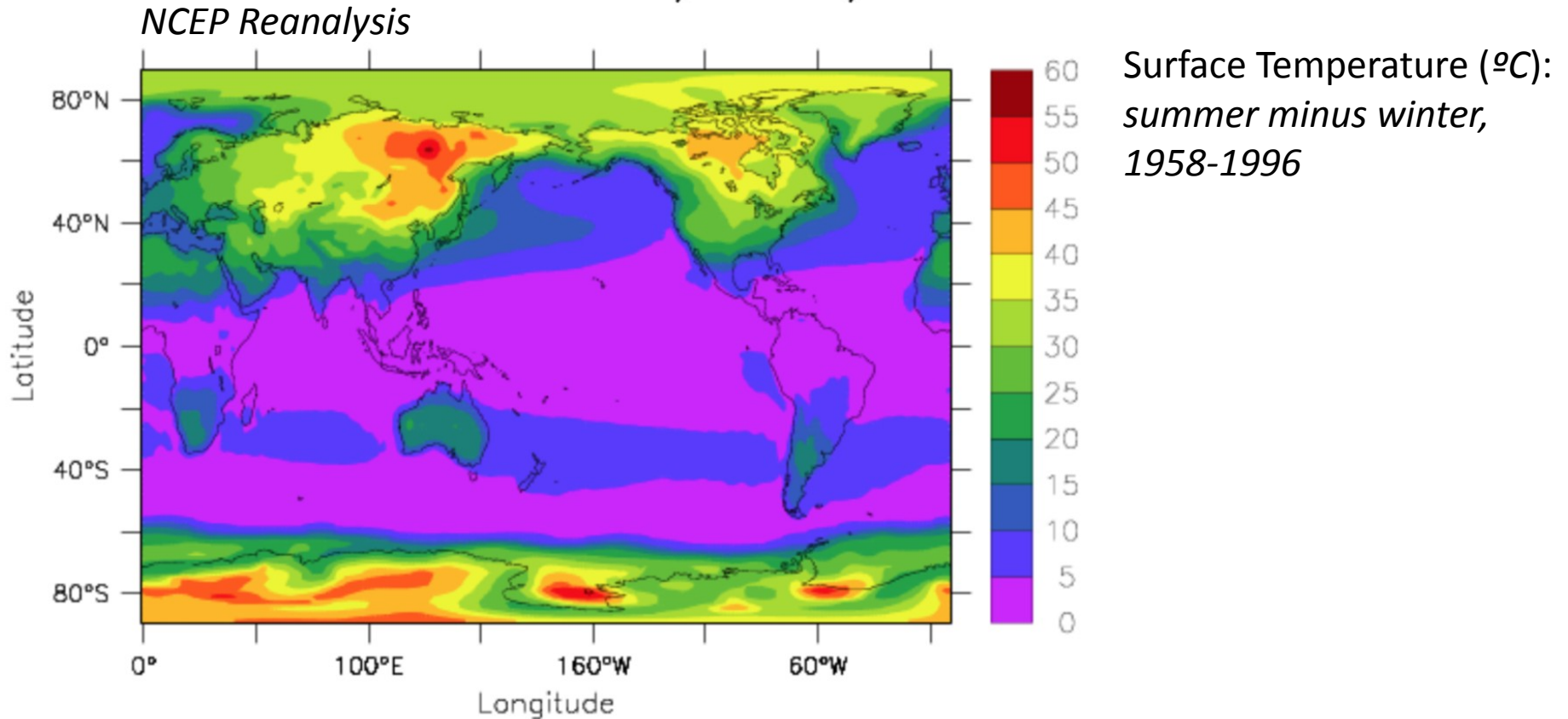
- Large heat capacity of the ocean

		Units	Air	Water
Specific Heat Capacity	c_p	J/(g°C)	1	4.2
Density	ρ	kg/m ³	1	1000
Volumetric Heat Capacity	$c_{vol} = c_p \rho$	J/(m ³ °C)	1000	4,200,000

- Volumetric heat capacity of water is **4200 times** that of air
- Top 2 m** of the ocean has the same heat capacity as the **whole** atmosphere.

- Land heat capacity << ocean heat capacity
→ temperature change over the ocean is slower than over land

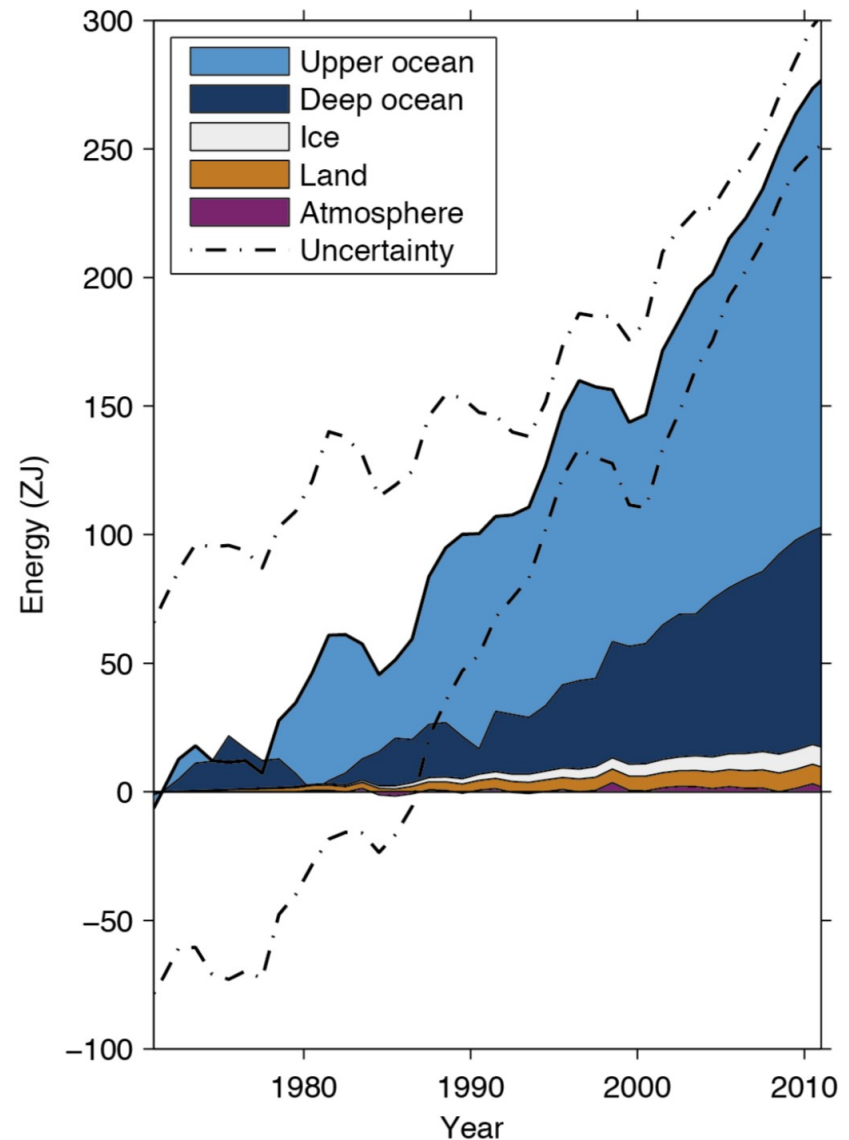
$$\Delta T / \Delta t = \Delta F / C.$$

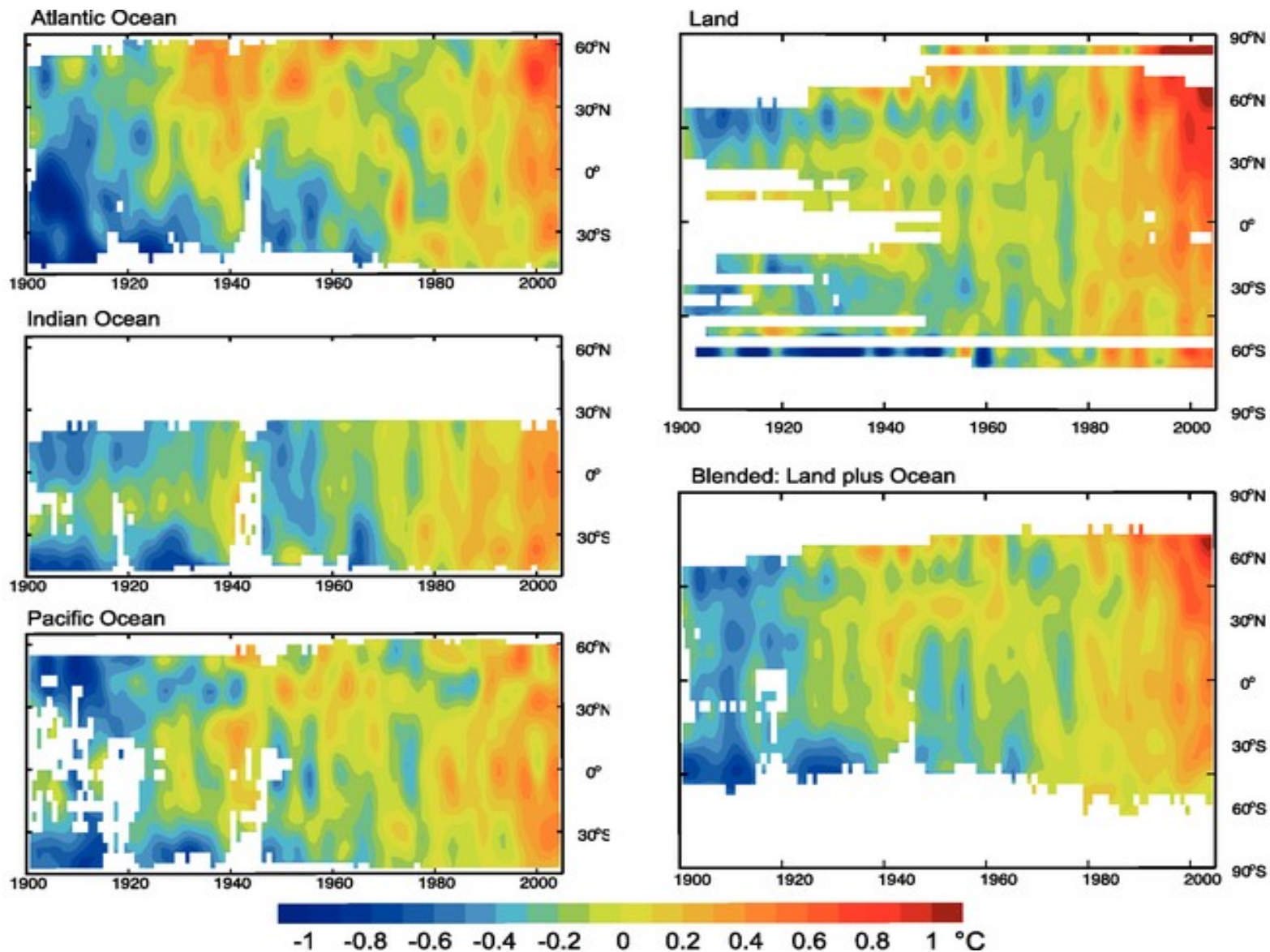


- The larger heat capacity of the ocean
→ dampens temperature variations on other timescales
→ warming is smaller over the ocean than over land

Heat Capacity

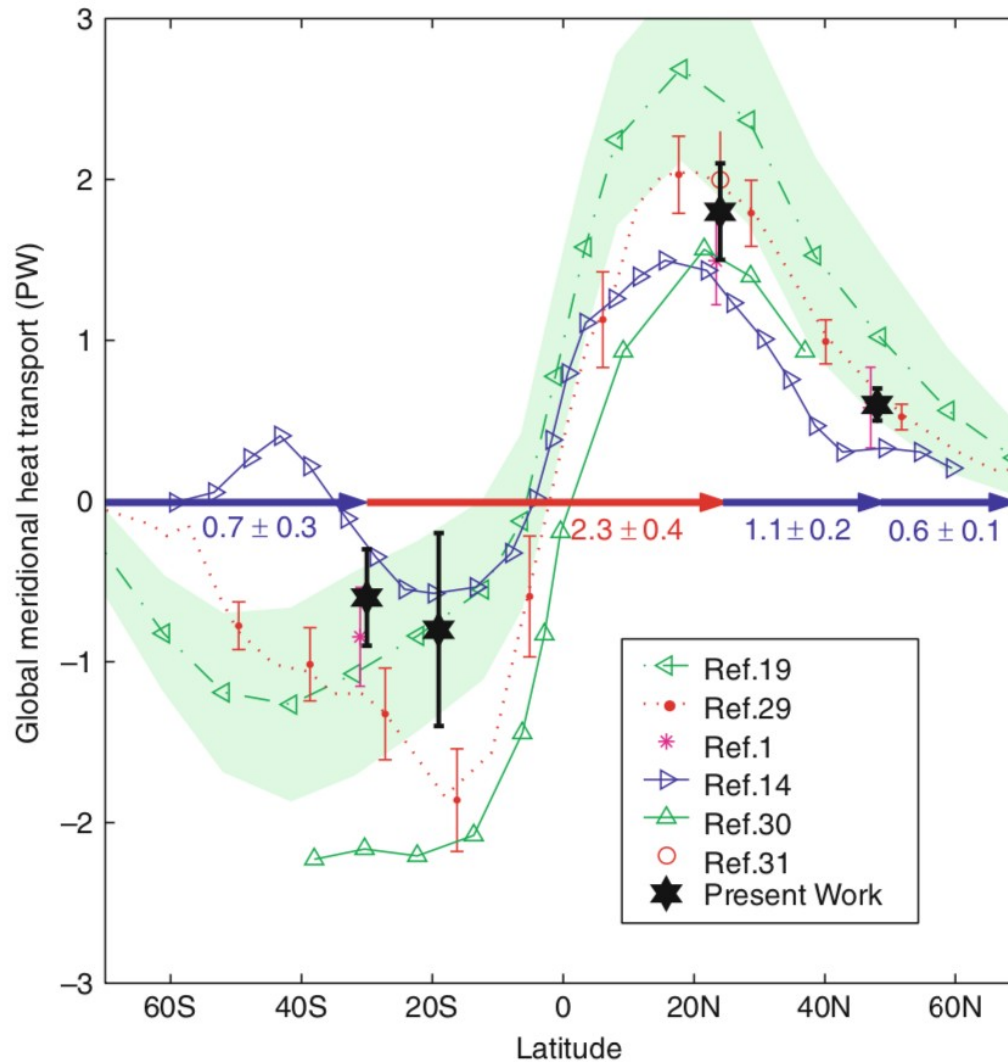
- ~90% of the observed increase in Earth's heat content goes into the **ocean**.
- Most of the remaining 10% goes into **ice** and **land**
- The heat gain of the **atmosphere** is very small.





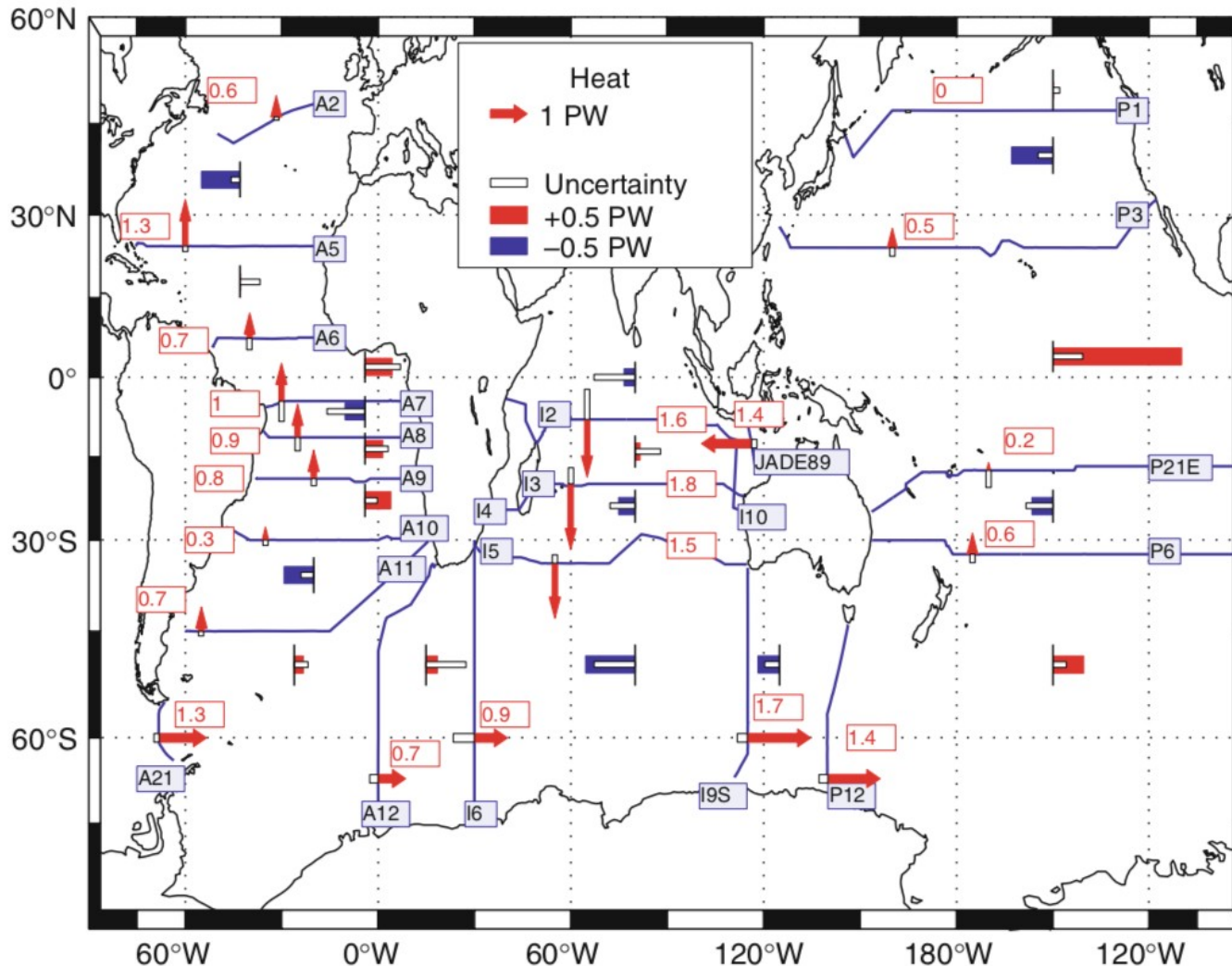
Zonal average of temperature anomaly compared to the period 1961-1990

Heat transport in the Ocean



Zonally integrated heat transport in the ocean based on observations and modelling (Ganachaud and Wunsch, 2000)

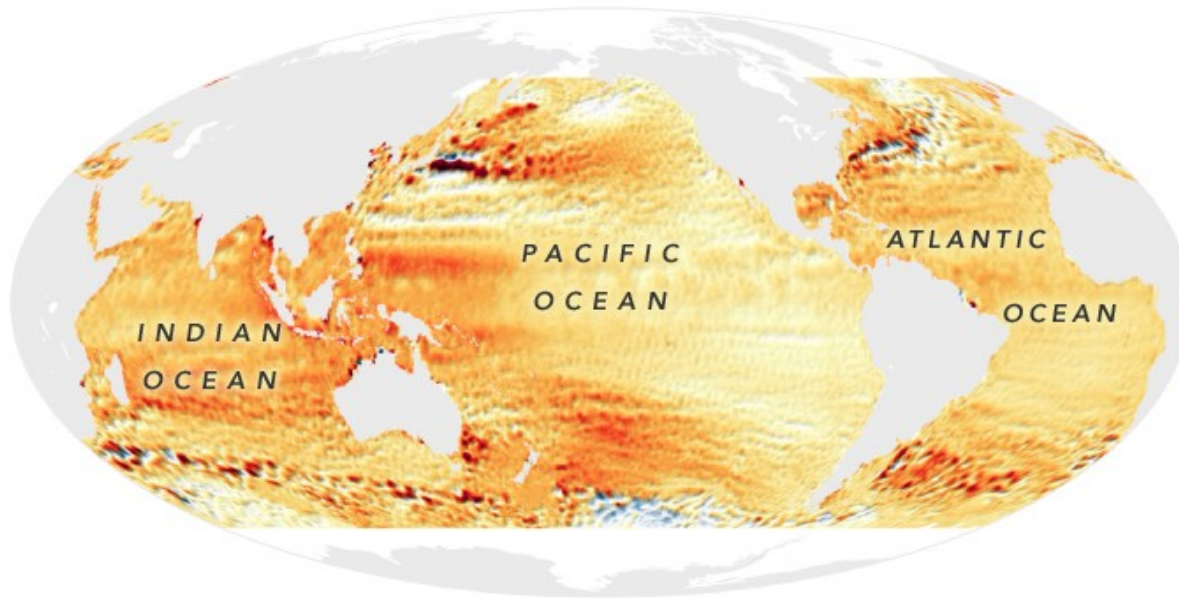
Heat transport in the Ocean



- Atlantic: Northward transports
- Indian Ocean: southward transport
- Pacific: surprisingly small
- Transport in the circumpolar current is largest with about 1.3 – 1.7 PW eastwards.

Meridional and vertical heat transports in the different regions of the world ocean. Numbers in boxes denote the meridional transport in PW. Horizontal bars represent the vertical transport (to the left – downwards) (Ganachaud and Wunsch, 2000)

Sea level rise



In-situ & satellite observations show that the overall level of the sea has been rising

1993-2018



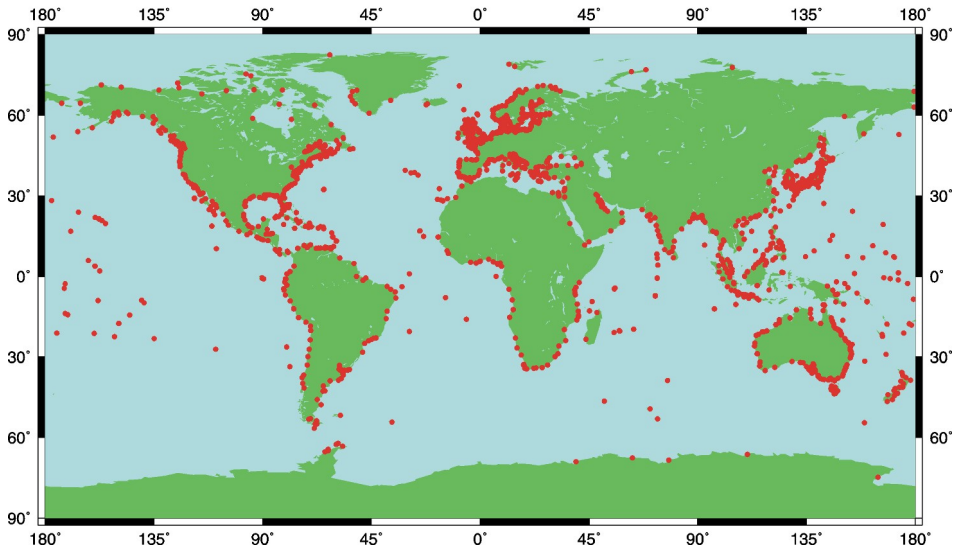
<https://earthobservatory.nasa.gov/>

Tide Gauges and Satellites Agree: Global Mean Sea Level is Rising
Observed change in sea level from 1996-2016 average (mm)

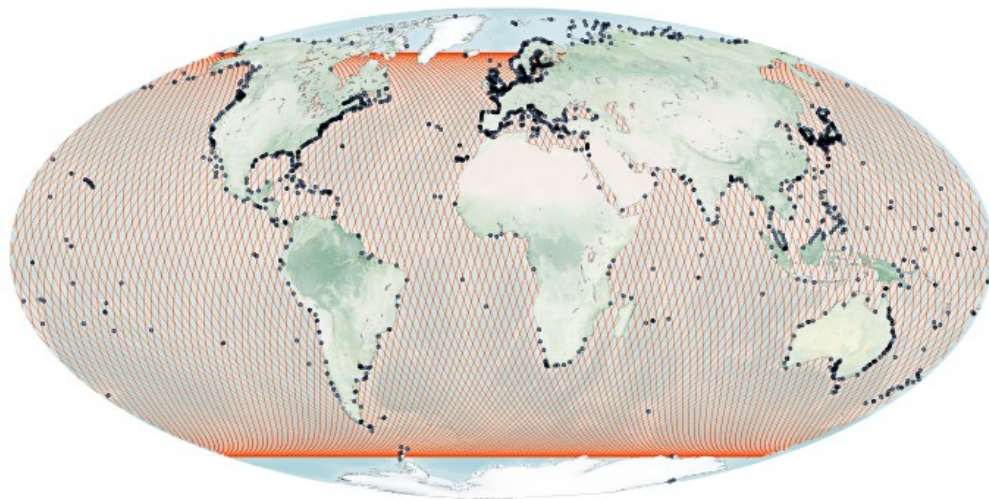


Sea level rise

Distribution of PSMSL Stations



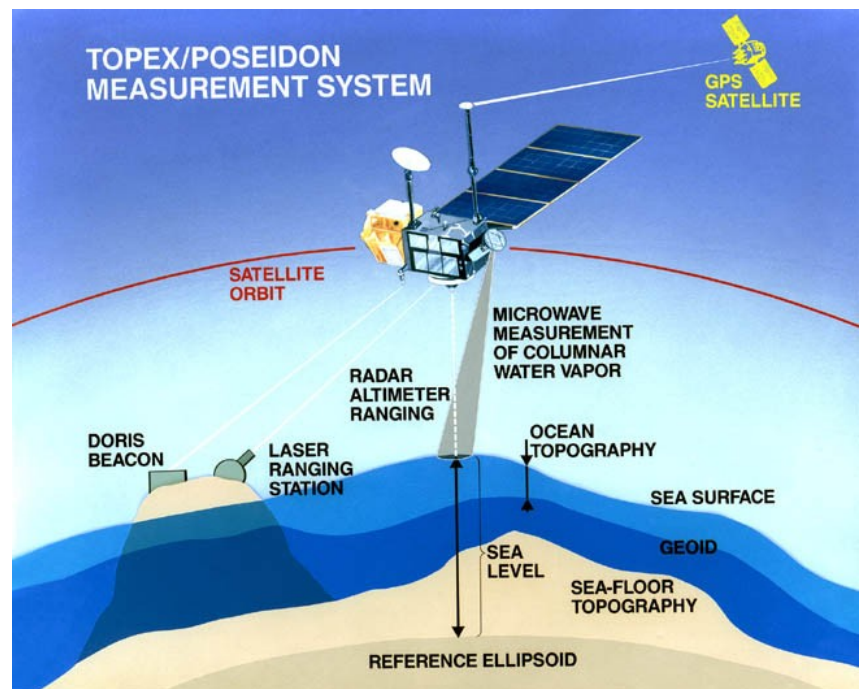
Satellite altimetry measurements cover significantly more area than tide gauges



• PSMSL Tide Gauges Jason-3 Orbits (10 days)

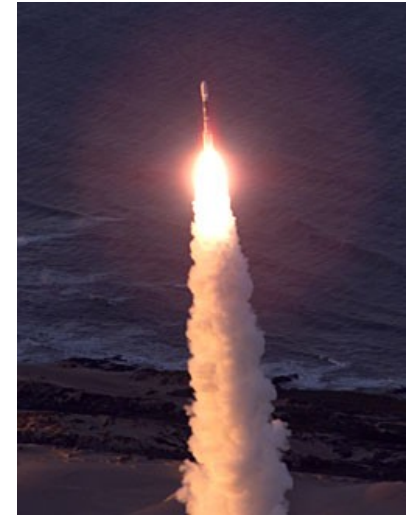
<https://earthobservatory.nasa.gov/>

satellite altimetry

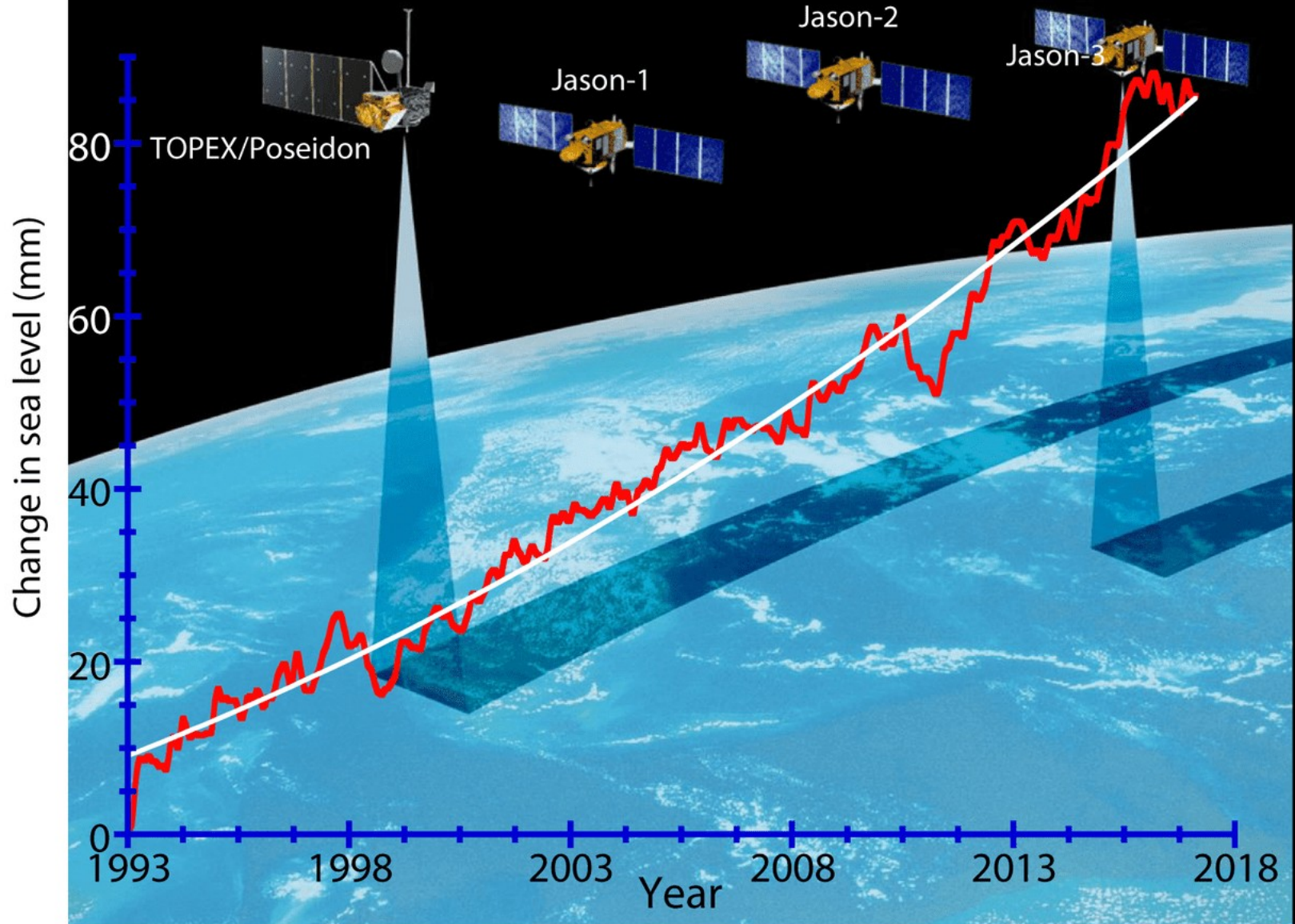


Past and Current altimeter satellites

Satellite	Years	Organisation	Accuracy
• SKYLAB	1972	NASA	20 m
• GEOS-3	1975-1978	NASA	3 m
• SEASAT	1978	NASA	2 m
• GEOSAT	1985-1990	US Navy	30 cm
• ERS-1	1991-1996	ESA	4-10 cm
• ERS-2	1995-2006	ESA	4 cm
• T/P	1992-2005	NASA/CNES	2 .. 3 cm
• GFO	2000-	US Navy	2 .. 5 cm
• JASON	2001-	NASA/CNES	2 .. 3 cm
• ENVISAT	2002-	ESA	2 .. 3 cm



Global Mean Sea Level Variations from Satellite Altimetry



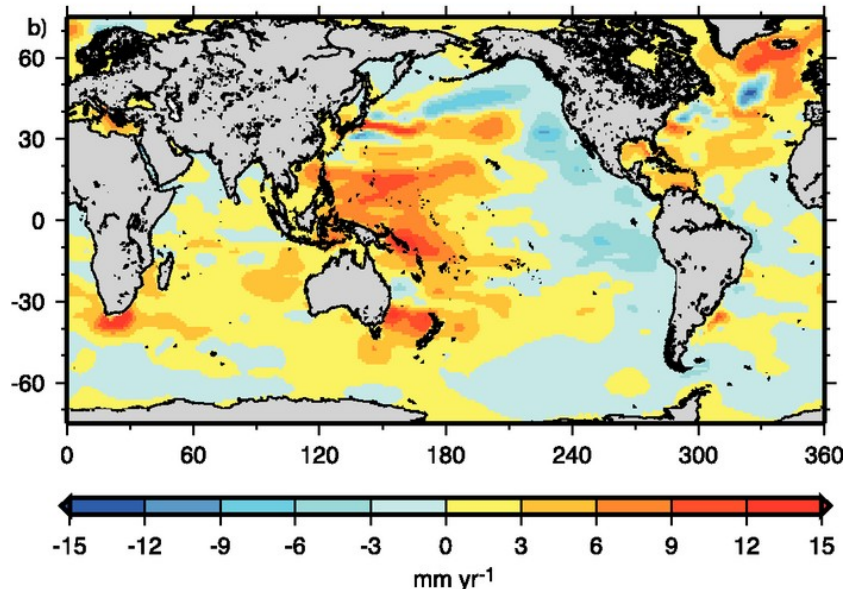
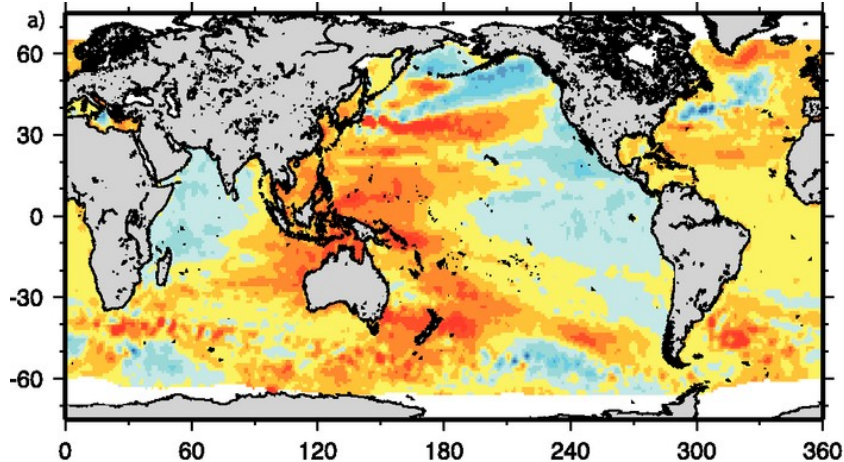
Satellite altimetry data. Illustration: John Fasullo

Sea level rise

- **Global Causes:**
 - Thermal expansion
 - Mass exchange with other reservoirs (Antarctica, Greenland, underground water in land, river systems, atmosphere, ..)
- **Local causes:**
 - Sinking land (groundwater withdrawal, tectonic plates)
 - Ocean circulation (e.g. the Gulf Stream is slowing, less water is taken from the East Coast of America → causing sea level to rise in places like Florida)

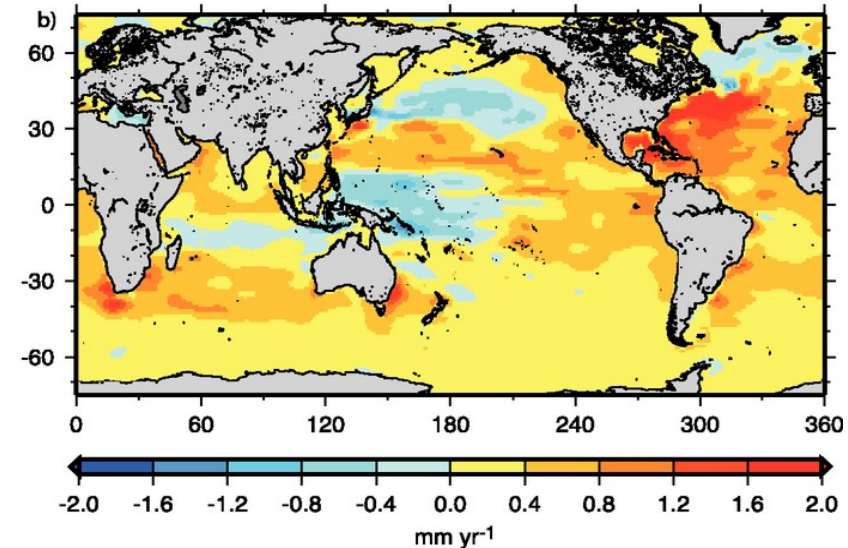
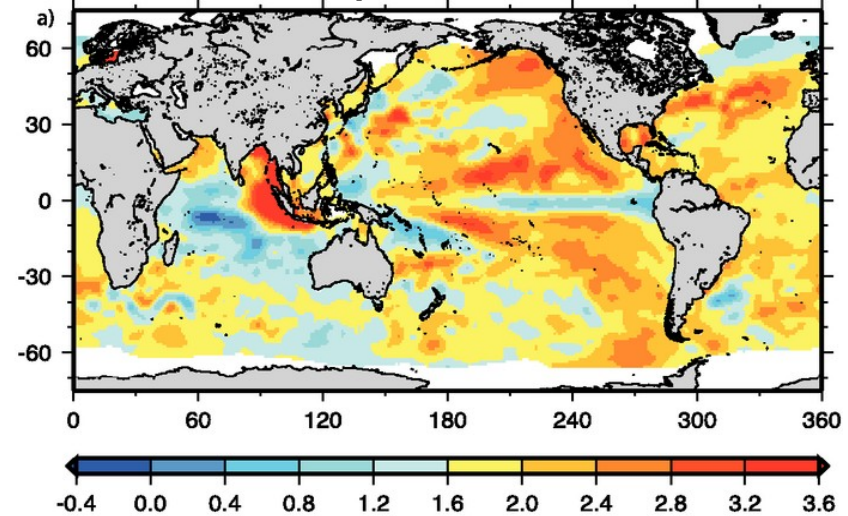
Sea level rise

1993-2003 Topex/Poseidon Sea level rise



Thermal expansion (0-700m)

1953-2003 Topex/Poseidon Sea level rise



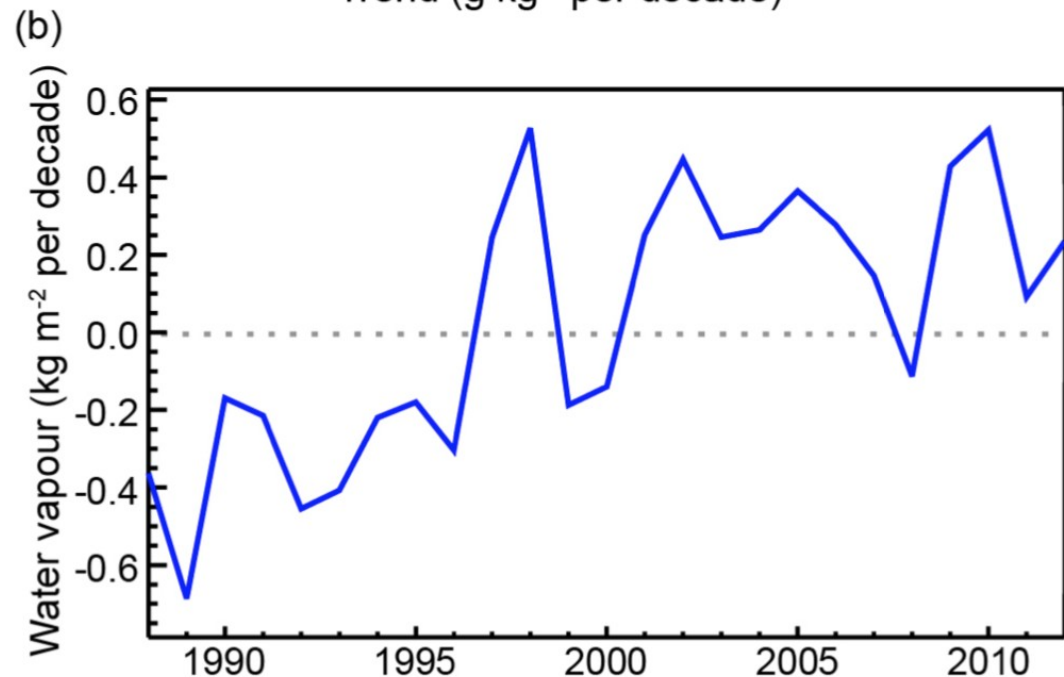
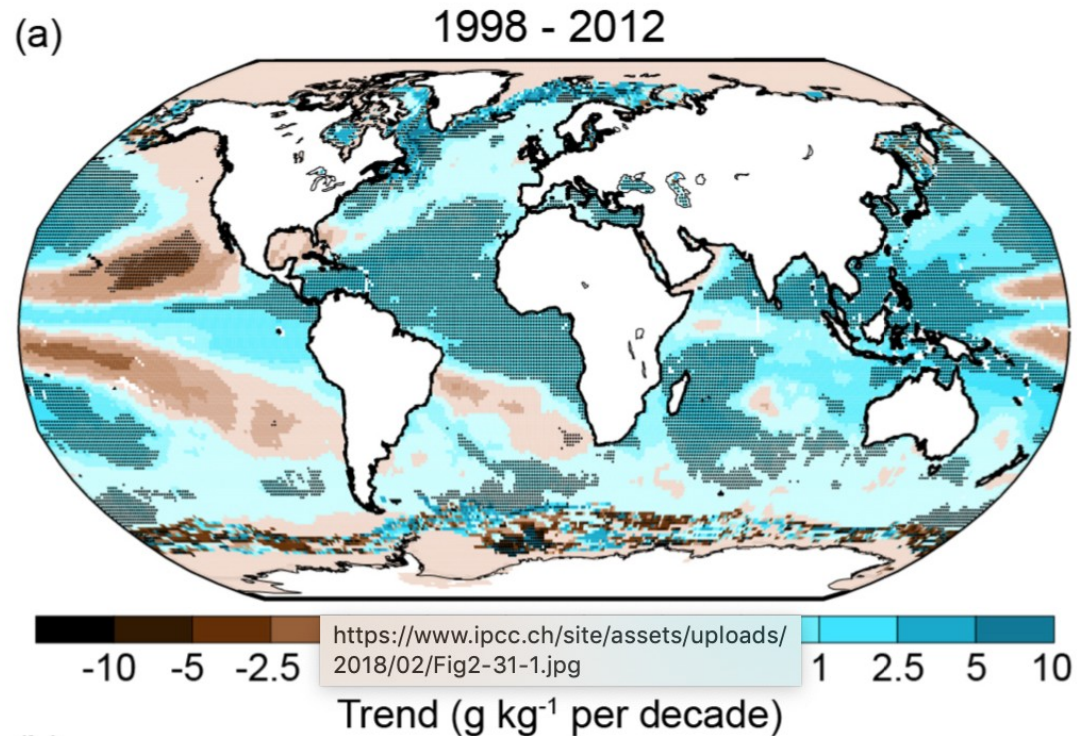
Thermal expansion (0-700m)

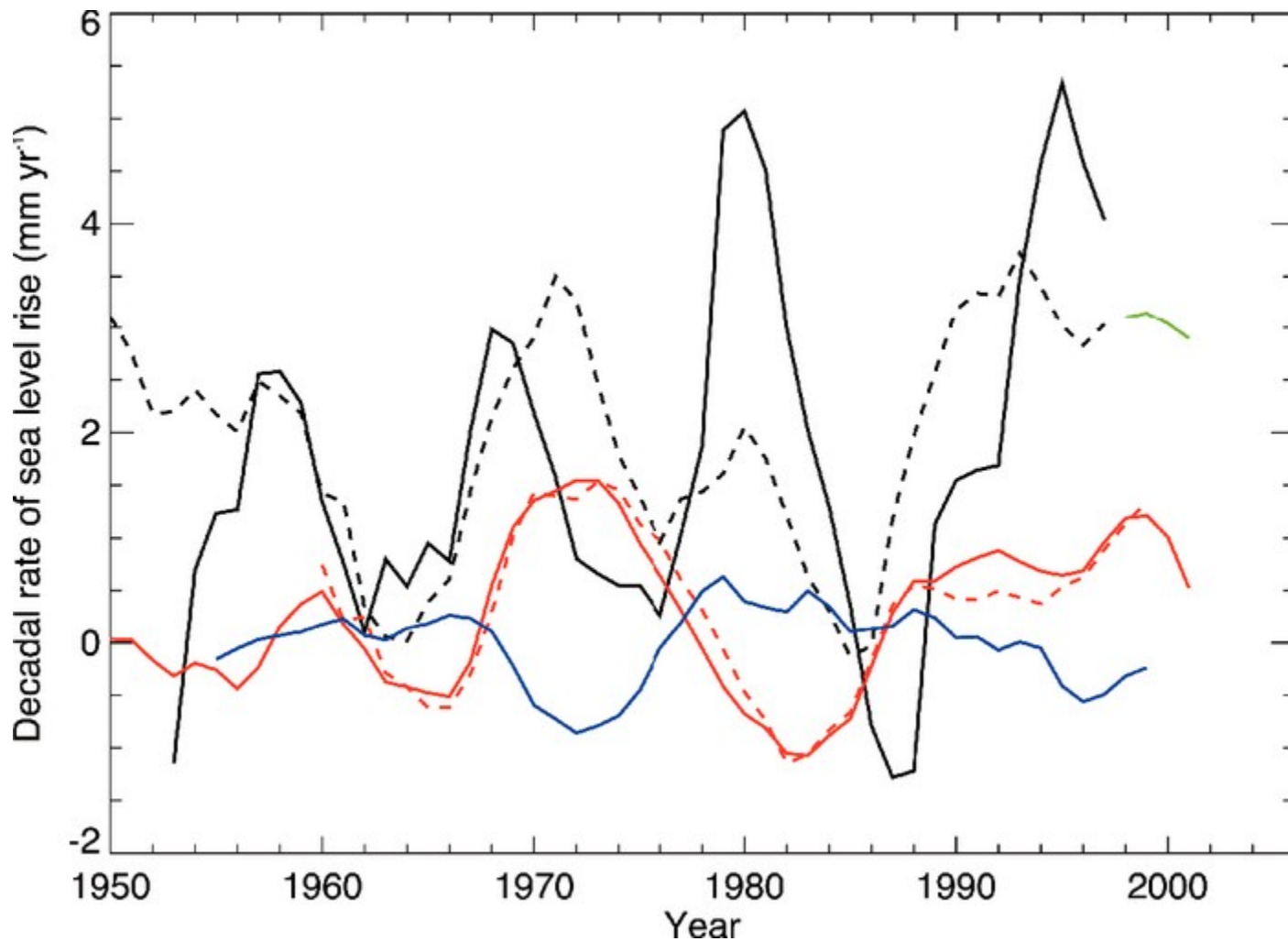
- increases in water vapor content in the atmosphere
- ← Warmer air can hold more water vapor

a) Observed changes in surface specific humidity estimated from satellites. Dark shading indicates significant changes.

a) Changes in globally averaged water vapor content.

From Harmann et al. (2013)





Overlapping 10-year rates of global sea level change from tide gauge data sets (Holgate and Woodworth, 2004, in solid black; Church and White, 2006, in dashed black) and satellite altimetry (updated from Cazenave and Nerem, 2004, in green), and contributions to global sea level change from thermal expansion (Ishii et al., 2006, in solid red; Antonov et al., 2005, in dashed red) and climate-driven land water storage (Ngo-Duc et al., 2005, in blue). Each rate is plotted against the middle of its 10-year period.

Practice Python #10.1

The CSV format

CSV (Comma Separated Values), e.g. sea level rise data from altimetry

Read and plot global mean sea level data observed by multiple altimetry missions

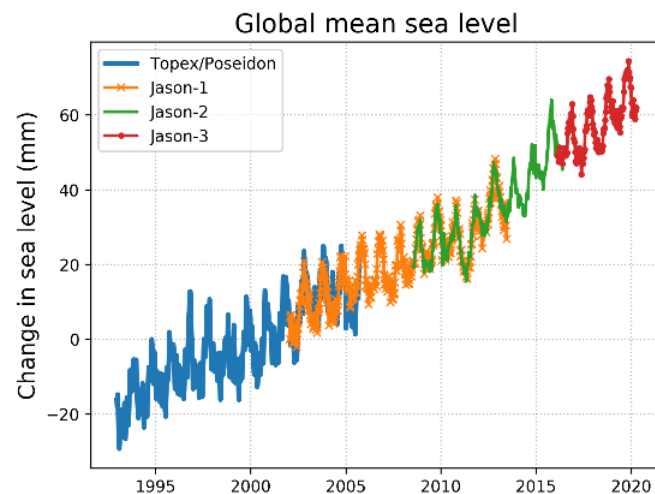
```
#title = mean sea level anomaly global ocean (66S to 66N) (Annual signals retained)
#trend = 2.94 mm/year (no glacial isostatic adjustment correction)
year,TOPEX/Poseidon,Jason-1,Jason-2,Jason-3
```

```
1992.9614,-16.24000,,,
1992.9865,-17.94000,,,
1993.0123,-14.84000,,,
1993.0407,-19.84000,,,
1993.0660,-25.24000,,,
1993.0974,-29.34000,,,
1993.1206,-27.64000,,,
1993.1493,-21.84000,,,
1993.1765,-18.94000,,,
1993.2037,-19.44000,,,
1993.2307,-22.94000,,,
1993.2851,-26.24000,,,
1993.3123,-20.04000,,,
1993.3394,-19.84000,,,

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<http://www.star.nesdis.noaa.gov/sod/lisa/SeaLevelRise>.

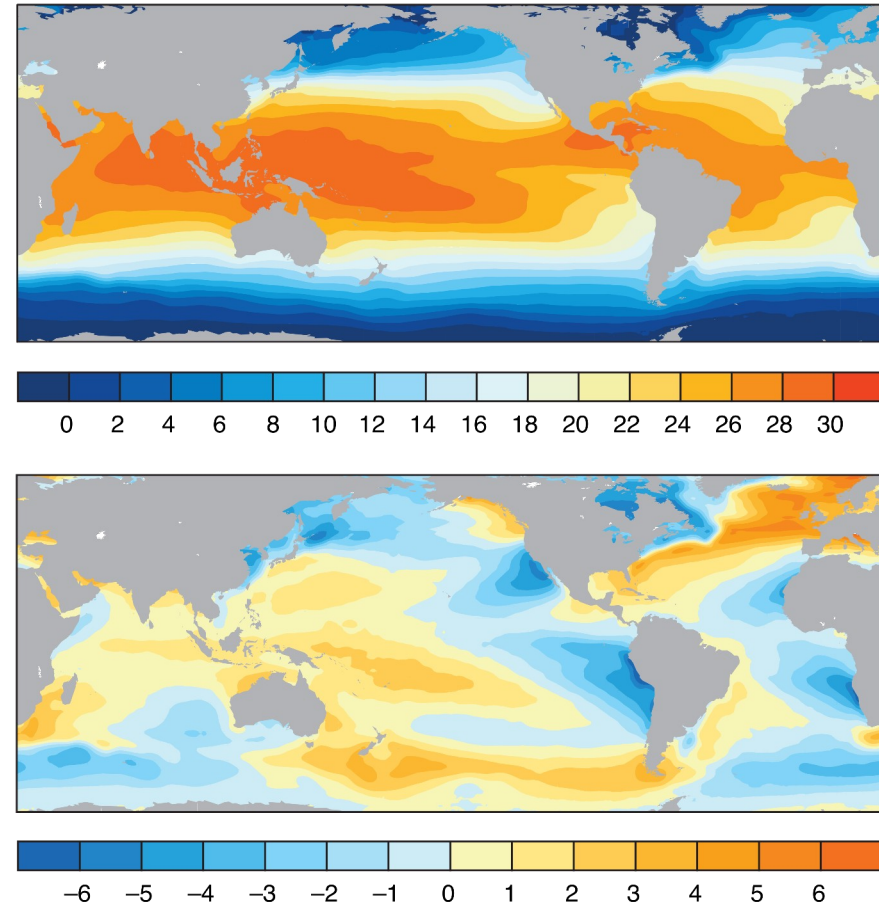
https://www.star.nesdis.noaa.gov/socd/lisa/SeaLevelRise/slr/slr_sla_gbl_keep_ref_90.csv



Practice #10.2: Sea Surface Temperature

SST netCDF data

1. Download sst.mnmean.nc from <https://www.esrl.noaa.gov/psd/data/gridded/data.noaa.oisst.v2.html>
2. Plot the average SST over the oceans (e.g. upper figure)
2. Plot the SST departures (e.g. lower figure)



https://www.star.nesdis.noaa.gov/socd/lsa/SeaLevelRise/slr/slr_sla_gbl_keep_ref_90.csv

Solution #1

```
import numpy as np
import matplotlib.pyplot as plt
data=np.genfromtxt("slr_sla_gbl_keep_txj1j2_90.csv",delimiter=',',comments='#')
time=data[:,0]
topex=data[:,1]
jason1=data[:,2]
jason2=data[:,3]
jason3=data[:,4]
plt.plot(time,topex,lw=3,label='Topex/Poseidon')
plt.plot(time,jason1,lw=3,label='Jason-1')
plt.plot(time,jason2,lw=3,label='Jason-2')
plt.plot(time,jason3,lw=3,label='Jason-3')
plt.legend(loc='upper left')
plt.ylabel('Sea level rise (mm)',fontsize=14)
plt.title('Global mean sea level',fontsize=16)
```

```
import pandas as pd
df =pd.read_csv(file,delimiter=',')
```

```
import numpy as np
import matplotlib.pyplot as plt
data=np.genfromtxt("slr_sla_gbl_keep_txj1j2_90.csv",delimiter=',',comments='#')
time=data[:,0]
topex=data[:,1]
jason1=data[:,2]
jason2=data[:,3]
jason3=data[:,4]
# to deal with NaN values
ttp=np.delete(time,np.argwhere(np.isnan(topex)))
tp=np.delete(topex,np.argwhere(np.isnan(topex)))
tjs1=np.delete(time,np.argwhere(np.isnan(jason1)))
js1=np.delete(jason1,np.argwhere(np.isnan(jason1)))
tjs2=np.delete(time,np.argwhere(np.isnan(jason2)))
js2=np.delete(jason2,np.argwhere(np.isnan(jason2)))
tjs3=np.delete(time,np.argwhere(np.isnan(jason3)))
js3=np.delete(jason3,np.argwhere(np.isnan(jason3)))
plt.plot(ttp,tp,lw=3,label='Topex/Poseidon')
plt.plot(tjs1,js1,lw=3,label='Jason-1')
plt.plot(tjs2,js2,lw=3,label='Jason-2')
plt.plot(tjs3,js3,lw=3,label='Jason-3')
plt.legend(loc='upper left')
plt.ylabel('Sea level rise (mm)',fontsize=14)
plt.title('Global mean sea level',fontsize=16)
```

Solution #2



Appendix. netCDF



- netCDF (network Common Data Form): a set of data formats, programming interfaces, and software libraries that help read and write scientific data files.
- developed and maintained at Unidata, University Corporation for Atmospheric Research (UCAR)

netCDF files

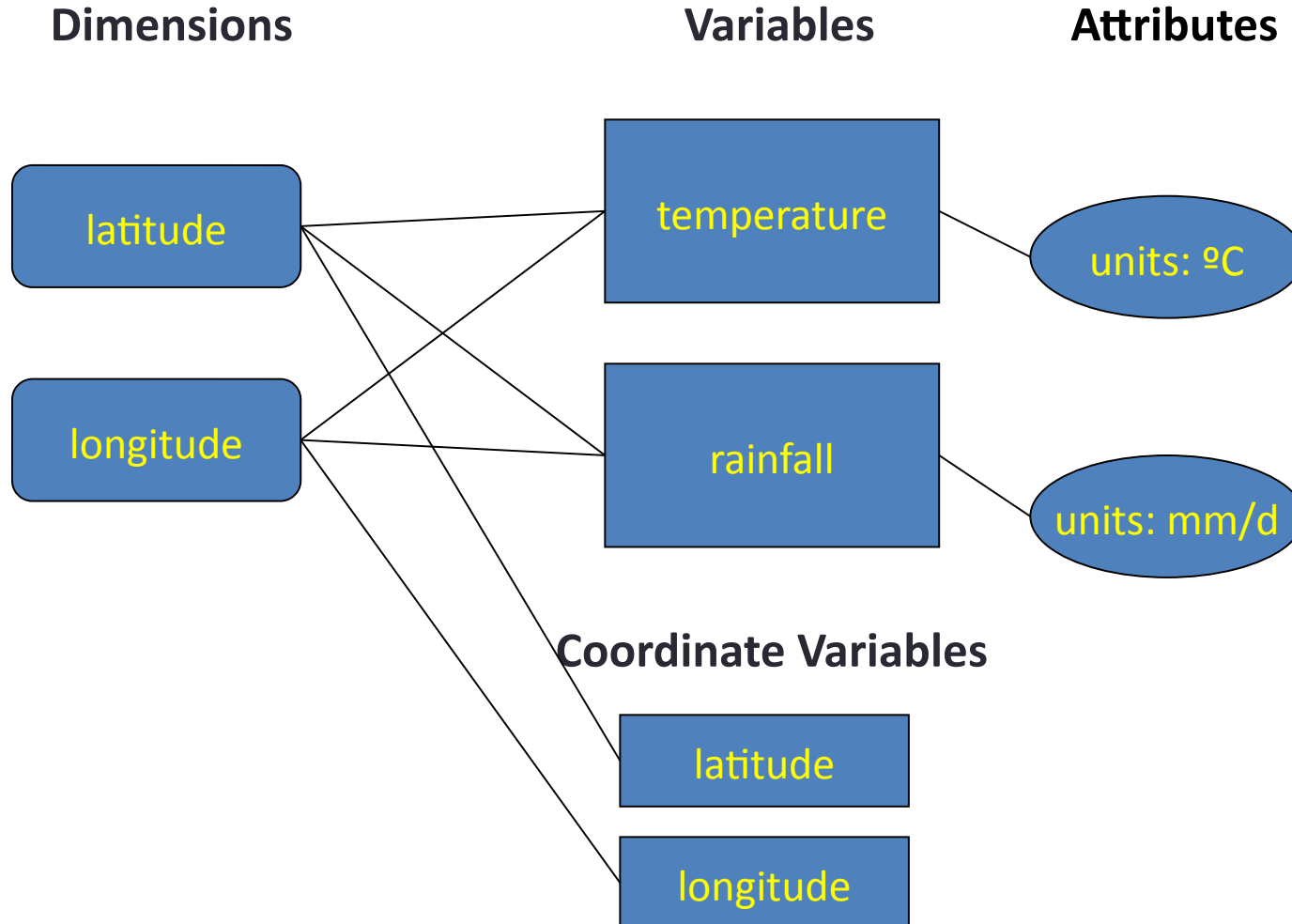
- **self describing** shows you what it contains
- **portable** [machine independent]

<http://www.unidata.ucar.edu/software/netcdf/>

Information normally contained in a netCDF file

- Dimensions
 - ✓ name, length
- Variables: named arrays
 - ✓ name, type, shape, attributes, values
 - **Fixed sized variables**: array of fixed dimensions
 - **Record variables**: array with its most-significant dimension **UNLIMITED** → **no limit in size, data can be extended along the unlimited dimension**
 - **Coordinate variables**: 1-D array with the same name as its dimension
- Attributes - “metadata about the data”
 - ✓ name, type, values, length
 - Variable attributes
 - Global attributes

netCDF file



How to examine a netCDF?

`Ncdump`, `ncgen` is Unidata utilities, available when you install netCDF

In Jupyter notebook

- `! ncdump file_name | less`
 - dumps the entire contents of a file
- `! ncdump -h file_name | less`
 - dumps the header info
- `! ncdump -v t2m file_name | less`

```
import numpy as np
import netCDF4 as nc
from netCDF4 import Dataset
# open a netcdf file
file='sst.mnmean.nc'
fh = Dataset(file, 'r') # Dataset is the class behavior to open the netCDF file
                        # fh means the file handle of the open netCDF file
# print fh format
print fh.file_format
# print info about dimensions
print fh.dimensions.keys()
print fh.dimensions['time']
# print info about variables
print fh.variables.keys()
# print attributes
print fh.Conventions
for attr in fh.ncattrs():
    print attr, '=', getattr(fh, attr)
# Extract fh from NetCDF file
lat = fh.variables['lat'][:] # extract/copy the fh
lon = fh.variables['lon'][:]
time = fh.variables['time'][:]
d_times=nc.num2date(fh.variables['time'],fh.variables['time'].units)
sst = fh.variables['sst'][:] # shape is time, lat, lon as shown above
sst_units=fh.variables['sst'].units
fh.close()
```